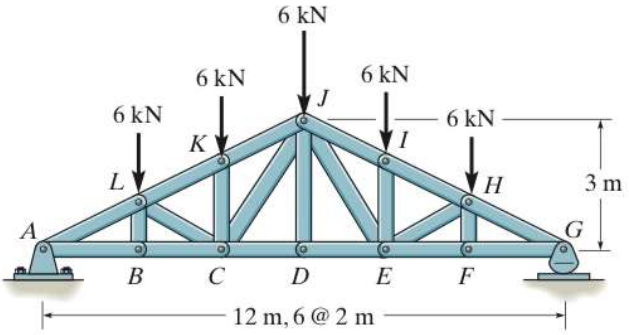
Faculty of Education
Dept. of Mathematics

Statics 1

MAT 117

Prepared by
Department of Mathematics

For first level students



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Chapter 1
Scalars & Vectors

1.1 Scalars and Vectors

Many physical quantities in engineering mechanics are measured using either scalars or vectors. A *scalar* is any positive or negative physical quantity that can be completely specified by its magnitude. Examples of scalar quantities include *length, mass,* and *time*. A *vector* is any physical quantity that requires both a magnitude and a direction for its complete description. Examples of vectors encountered in statics are force, position, and moment. A vector is shown graphically by an arrow. The length of the arrow represents the magnitude of the vector, and the angle θ between the vector and fixed axis defines its direction. Every vector in mechanics is well visualized as an arrow. The direction of the arrow is the direction of the vector (see fig.1.1). The length of the arrow is proportional to the magnitude of the vector. The magnitude of $\vec{A}$ is a positive scalar indicated by $|\vec{A}|$.

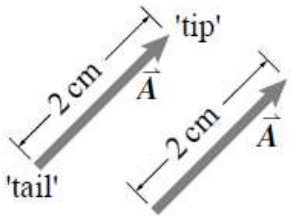


Fig. 1.1

You must use a notation that distinguishes between vectors and scalars using, for example, $\vec{a}$, or $\underline{a}$ for acceleration and a or $|\vec{a}|$ for the magnitude of acceleration.

1.2. Vector Operations

1.2.1 Adding vectors

The sum of two vectors $\vec{A}$ and $\vec{B}$ is defined by the *tip to tail* rule of vector addition shown in Fig. 1.2 for the sum

$$\vec{C} = \vec{A} + \vec{B}.$$

Vector $\vec{A}$ is drawn. Then vector $\vec{B}$ is drawn with its tail at the tip (or head) of $\vec{A}$. The sum $\vec{C}$ is the vector from the tail of $\vec{A}$ *to the tip of* $\vec{B}$. Also, Fig. 1.2 shows that, if two vectors $\vec{A}$ and $\vec{B}$, treated as free vectors, may be replaced by their equivalent vector $\vec{C}$, which is the diagonal of the parallelogram formed by $\vec{A}$ and $\vec{B}$ as its two sides.

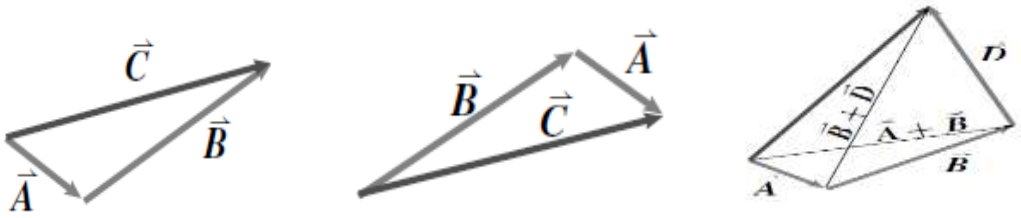


Fig. 1.2

- As a special case, if the two vectors $\vec{A}$ and $\vec{B}$ are collinear, i.e., both have the same line of action, the parallelogram law reduces to an algebraic or scalar addition C = A + B, as shown in Fig. 1.3.

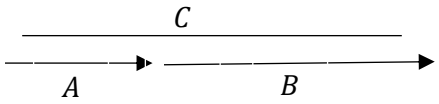


Fig. 1.3

1.2.2. Adding many vectors

We can generalize the previous rule to add two vectors. As we can observe from the following figure (Fig. 1.4), the vector $\vec{AE}$ is a yield vector or the net vector for the vectors $\vec{AB}, \vec{BC}, \vec{CD}, \vec{DE}$. So, we have.

$$\vec{AB} + \vec{BC} + \vec{CD} + \vec{DE} = \vec{AE}$$

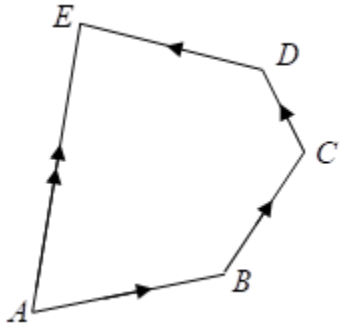


Fig. 1.4

1.2.3. Subtraction vectors

Subtraction is most simply defined by inverse addition. Find $\vec{C} - \vec{A}$ means find the vector which when added to $\vec{A}$ gives $\vec{C}$. We can draw $\vec{C}$, draw $\vec{A}$ and then find the vector which,

when added tip to tail to $\vec{A}$ give $\vec{C}$. The resultant of the difference between two vectors A and B of the same type may be expressed as:

$$\vec{R'} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B}).$$

This vector sum is shown graphically in Fig. 1.5. Subtraction is therefore defined as a special case of addition, so the rules of vector addition also apply to vector subtraction.

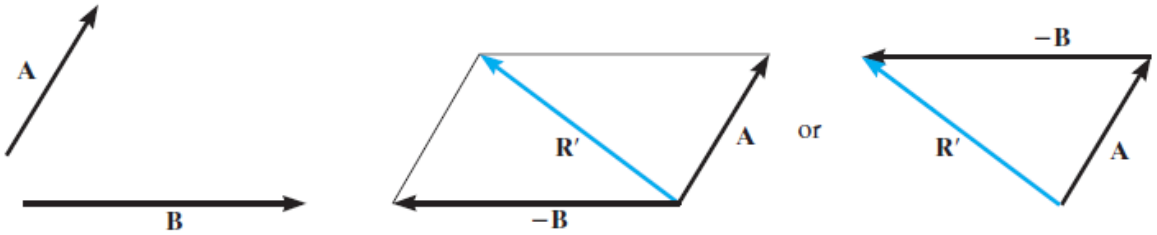


Fig. 1.5

- Zero vector is obtained when the vector is subtracted from itself; i.e., $\vec{A} - \vec{A} = 0$. this is also called a *null vector*.

1.3. Multiplication of vector by scalars

The product of a vector $\vec{A}$ and a scalar m is a vector $m\vec{A}$ whose magnitude is m times the magnitude of $\vec{A}$ and similarly or oppositely the direction of $\vec{A}$, depending on whether m is positive or negative.

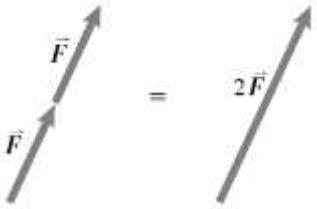


Fig. 1.6

- Note that, $c(\vec{A} + \vec{B}) = c\vec{A} + c\vec{B}$

1.4. Relative position vectors

The concept of relative position permeates most mechanics equations. The position of point B relative to point A is represented by the vector $\vec{r}_{B/A}$ (it may be written $\vec{r}_{AB}$) which can be obviously drawn from A and to B (as shown in Fig. 1.7). Fig. 1.7 shows that the relative positions add by the tip to tail rule. That is,

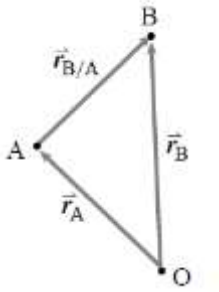


Fig. 1.7

$$\vec{r}_{B/A} = \vec{r}_B - \vec{r}_A$$

1.5. Unit vectors

Unit vectors are vectors with a magnitude of one. Unit vectors are useful for indicating direction. Key examples are the unit vectors pointed in the positive *x; y* and *z* directions $\hat{\imath}, \hat{\jmath}$ and $\hat{k}$ (these unit vectors are called orthogonal unit vectors). An easy way to find a unit vector in the direction of a vector $\vec{A}$ is to divide $\vec{A}$. by its magnitude $|\vec{A}|$, thus $\hat{A} = \vec{e_A} = \frac{\vec{A}}{|\vec{A}|}$.

A common situation is to know that a vector $\vec{R}$ is a yet unknown scalar *R* multiplied by a unit vector pointing between known points *A* and *B*. We can then write

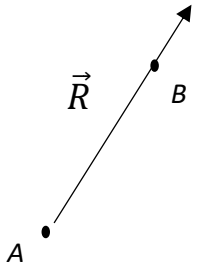


Fig. 1.8

$$\vec{R} = \left|\vec{R}\right| \hat{R} = R \frac{\vec{r_B} - \vec{r_A}}{\left|\vec{r_B} - \vec{r_A}\right|}$$

where we have used $\hat{R}$ as the unit vector pointing from *A* to *B* (as seen in Fig. 1.8).

1.6. The components of a vectors

A given vector $\vec{F}$, can be described as the sum of vectors each of which is parallel to a coordinate axis. Thus the vector $\vec{F}$ in 3D can be written as the product of a scalar and a unit vector along the positive axes (Fig. 1.9). So

$$\vec{F} = F_x \underline{i} + F_y \underline{j} \qquad \text{(2D)}$$

$$\vec{F} = F_x \underline{i} + F_y \underline{j} + F_z \underline{k} \qquad \text{(3D)}$$

Then the magnitude of a vector $\vec{F}$ can be described as

$$\left|\vec{F}\right| = \sqrt{{F_x}^2 + {F_y}^2} \quad or \quad \left|\vec{F}\right| = \sqrt{{F_x}^2 + {F_y}^2 + {F_z}^2}$$



Fig. 1.9

Example1.1 The velocity of a car is given $\vec{v} = 30\,\hat{\imath} + 40\,\hat{\jmath}\ m/sec$

(a) Find the speed (magnitude of $\vec{v}$) of the car

(b) Find a unit vector in the direction of $\vec{v}$

(c) Write the velocity vector as a product of its magnitude and the unit vector.

Solution

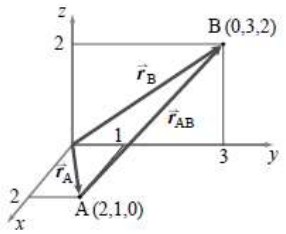
(a) speed$= v = \left|\vec{v}\right| = \sqrt{v_x^2 + v_y^2} = \sqrt{30^2 + 40^2} = 50\ m/sec$

(b) $\hat{v} = \frac{\vec{v}}{\left|\vec{v}\right|} = \frac{30\,\hat{\imath}+40\,\hat{\jmath}}{|30\,\hat{\imath}+40\,\hat{\jmath}|} = \frac{30\,\hat{\imath}+40\,\hat{\jmath}}{\sqrt{30^2+40^2}} = \frac{3}{5}\,\hat{\imath} + \frac{4}{5}\,\hat{\jmath}$ (it has no units)

(b) $\vec{v} = \left|\vec{v}\right|\ \hat{v} = 50\left(\frac{3}{5}\,\hat{\imath} + \frac{4}{5}\,\hat{\jmath}\right).\ m/sec$

Example1.2 Let $A\ (2m, 1m, 0)$ and $B\ (0, 3m, 2m)$ be two points in the *xyz* coordinate system. Find the position vector of point *B* with respect to point *A*.

Solution



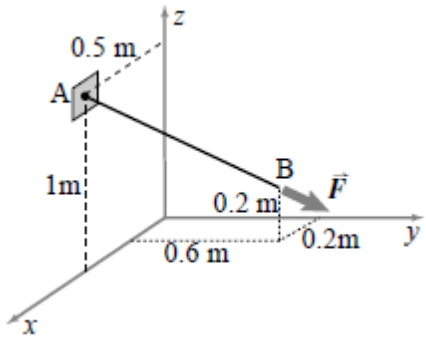
$$\vec{r}_{AB} = \vec{r}_B - \vec{r}_A$$

$$= \left(0\,\hat{\imath} + 3\,\hat{\jmath} + 2\,\hat{k}\right) - \left(2\,\hat{\imath} + 1\,\hat{\jmath} + 0\,\hat{k}\right)$$

$$= -2\,\hat{\imath} + 2\,\hat{\jmath} + 2\,\hat{k} = 2\left(-\,\hat{\imath} + \,\hat{\jmath} + \,\hat{k}\right)m$$

Example1.3 A string is pulled with a force *F =100N* as shown in the Fig. 1.10. Write *F* as a vector.

Solution



$$\vec{A} = 0.5\,\hat{\imath} + \,\hat{k}, \quad \vec{B} = 0.2\,\hat{\imath} + 0.6\,\hat{\jmath} + 0.2\,\hat{k}$$

$$\vec{r}_{AB} = \vec{r}_B - \vec{r}_A$$

$$\vec{r}_{AB} = -0.3\,\hat{\imath} + 0.6\,\hat{\jmath} - 0.8\,\hat{k}$$

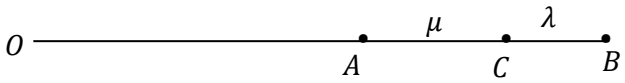
$$\vec{F} = F\frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} = 100\frac{-0.3\,\hat{\imath} + 0.6\,\hat{\jmath} - 0.8\,\hat{k}}{\sqrt{(-0.3)^2 + (0.6)^2 + (-0.8)^2}}$$

$$\vec{F} = -0.29\,\hat{\imath} + 0.57\,\hat{\jmath} - 0.77\,\hat{k}$$

Theorem: If AB is a straight line, C is a point on it such that $\lambda \underline{AC} = \mu \underline{CB}$, then

$$\lambda\ \underline{OA} + \mu\ \underline{OB} = (\lambda + \mu)\ \underline{OC}$$

where o is an any point.



Proof:

(i) Let o exist on the line AB or on its direction, so we have

$$\lambda\ OA + \mu\ OB = \lambda(OC - OC) + \mu(OC + CB)$$
$$= (\lambda + \mu)OC - \lambda AC + \mu CB$$

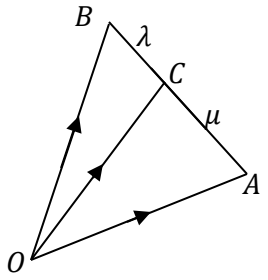
But $$\lambda\ AC = \mu\ CB$$

$$\therefore \lambda\ OA + \mu\ OB = (\lambda + \mu)OC$$ # Q.E.D

(ii) Let O exist out the line AB as we can note from the following figure, so we have

$$\underline{OA} = \underline{OC} + \underline{CA}$$

$$\underline{OB} = \underline{OC} + \underline{CB}$$



By multiplying the first equation by λ and the second equation by μ and adding the result we have:

$$\lambda\ \underline{OA} + \mu\ \underline{OB} = \lambda(\underline{OC} + AC) + \mu(\underline{OC} + \underline{CB})$$
$$= (\lambda + \mu)\underline{OC} + \lambda\underline{CA} + \mu\underline{CB}$$

But $$\lambda\ \underline{AC} = \mu\ \underline{CB}$$

$$\therefore \lambda\ \underline{CA} + \mu\ \underline{CB} = \lambda\ \underline{CA} + \mu\ \underline{AC} = 0$$
$$\therefore \lambda\ \underline{OA} + \mu\ \underline{OB} = (\lambda\ + \mu)\ \underline{OC}$$

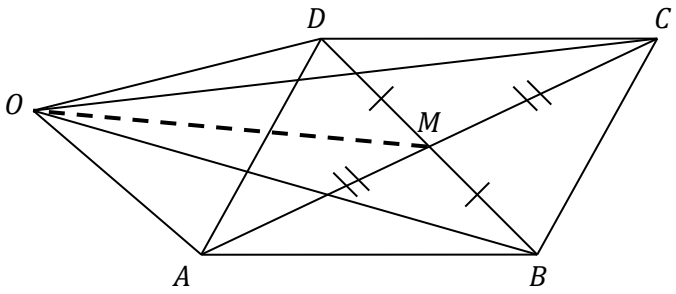
Special case If $\lambda = \mu = 1$, so we have $\underline{OA} + \underline{OB} = 2\ \underline{OC}$

Example1.4 If O is a point on the plane of the parallelogram $ABCD$and M is an intersection point of AC and BD. Prove that

$$\underline{OA}\ +\ \underline{OB}\ +\ \underline{OC}\ +\ \underline{OD} = 4\ \underline{OM}$$

Solution

We must note that in the triangle OAC, where OM is a median line in it so, we have:



$$\therefore \ \underline{OA} + \underline{OC} = 2\ \underline{OM} \qquad (1)$$

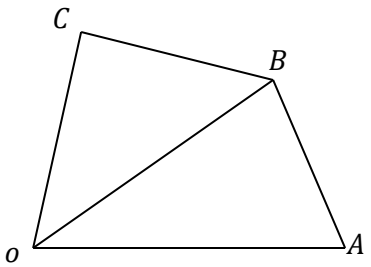
Also, must observe that in the triangle OBD, where oM is a median line in it so, we have:

$$\therefore \ \underline{OB} + \underline{OD} = 2\ \underline{OM} \qquad (2)$$

After adding Eqs. (1) and (2) we get the following required relation

$$\underline{OA} + \underline{OB} + \underline{OC} + \underline{OD} = 4\ \underline{OM}$$

Example1.5 If the two following conditions are satisfying $\lambda\underline{oA} + \mu\underline{oB} + \nu\underline{oC} = 0$ and $\lambda + \mu + \nu = 0$, where λ, μ, ν are constants. Prove that the three points A, B, C must be collinear.



Solution

We assume the A, B, C does not located at same straight-line as shown in this figure

$$\because \ \underline{oA} = \underline{oB} + \underline{BA}\ ,$$
$$\underline{oC} = \underline{oB} + \underline{BC}$$

By using the first condition $\lambda\underline{oA} + \mu\underline{oB} + \nu\underline{oC} = 0$, we note that:

$$\lambda(\underline{oB} + \underline{BA}) + \mu\underline{oB} + \nu(\underline{oB} + \underline{BC}) = 0$$

$$\therefore (\lambda + \mu + \nu)\underline{oB} + \lambda\underline{BA} + \nu\underline{BC} = 0$$

Also by using the second condition $\lambda + \mu + \nu = 0$, we observe that:

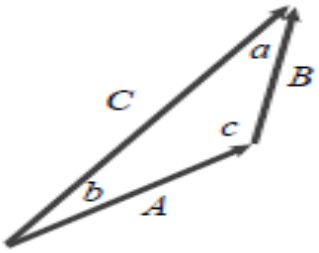
$$\therefore \lambda\underline{BA} + \nu\underline{BC} = 0$$

$$\therefore \quad \nu\underline{BC} = -\lambda\underline{BA} = \lambda\underline{AB}$$

The last relation means the three points A, B, C are collinear.

1.7 Vector triangles and the laws of sines and cosines

The tip to tail rule of vector addition defines a triangle. Given some information about the vectors in this triangle how does one figure out the rest? One traditional approach is to use the laws of sines and cosines.

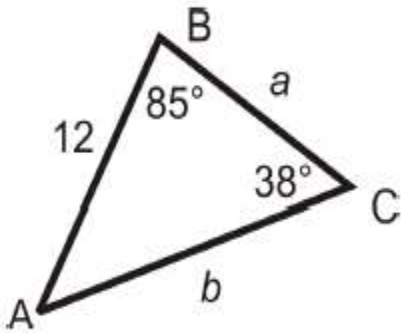


Consider the vector sum $\underline{A} + \underline{B} = \underline{C}$ represented by the triangle shown with traditionally labeled sides A; B; and C and internal angles a; b, and c. So the sides and angles are related by the following laws:

(i) $\frac{\sin a}{A} = \frac{\sin b}{B} = \frac{\sin c}{C}$ which represents the law of sines.

(ii) $C^2 = A^2 + B^2 - 2AB\cos c$ (The law of cosines).

Example1.6. Solve the following triangle by using the Law of Sines.



Solution

First, to find $m\angle A$, we can use the Triangle Sum Theorem.

$$m\angle A + 85° + 38° = 180°$$
$$\Rightarrow m\angle A = 57°$$

Now, use the Law of Sines to set up ratios for a and b.

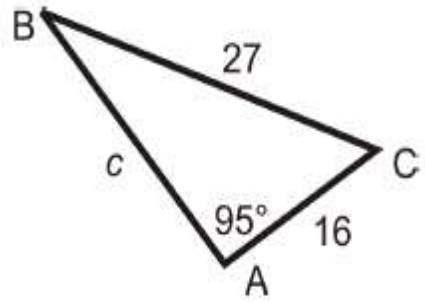
$$\frac{\sin 57°}{a} = \frac{\sin 85°}{b} = \frac{\sin 38°}{12}$$

$$\frac{\sin 57°}{a} = \frac{\sin 38°}{12}$$
$$a \cdot \sin 38° = 12 \cdot \sin 57°$$
$$a = \frac{12 \cdot \sin 57°}{\sin 38°} \approx 16.4$$

$$\frac{\sin 85°}{b} = \frac{\sin 38°}{12}$$
$$b \cdot \sin 38° = 12 \cdot \sin 85°$$
$$b = \frac{12 \cdot \sin 85°}{\sin 38°} \approx 19.4$$

Example1.7. Solve the following triangle by using the Law of Sines.

Solution:



Set up the ratio for $m\angle B$ using Law of Sines.

$$\frac{\sin 95^\circ}{27} = \frac{\sin B}{16}$$
$$27 \cdot \sin B = 16 \cdot \sin 95^\circ$$
$$\sin B = \frac{16 \cdot \sin 95^\circ}{27} \rightarrow \sin^{-1}\left(\frac{16 \cdot \sin 95^\circ}{27}\right) = 36.2^\circ$$

To find $m\angle C$, ust the triangle angle sum theorem

$$m\angle C + 95^\circ + 36.2^\circ = 180^\circ$$
$$\Rightarrow m\angle C = 48.8^\circ$$

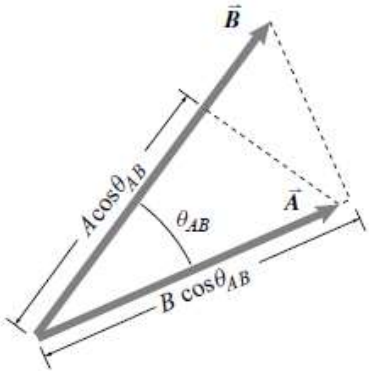
To find c, use the law of sines again $\frac{sin\,95^\circ}{27} = \frac{sin\,48.8^\circ}{c}$

$$\frac{\sin 95^\circ}{27} = \frac{\sin 48.8^\circ}{c}$$

$$c \cdot \sin 95^\circ = 27 \cdot \sin 48.8^\circ$$
$$c = \frac{27 \cdot \sin 48.8^\circ}{\sin 95^\circ} \approx 20.4$$

1.8 Dot or Scalar product of two vectors

The dot product is used to project a vector in a given direction, to reduce a vector to components, to reduce vector equations to scalar equations, to define work and power, and to help solve geometry problems. The *dot product* of two vectors $\underline{A}$ and $\underline{B}$ is written $\underline{A}.\underline{B}$ (pronounced 'A dot B'). The dot product of $\underline{A}$ and $\underline{B}$ is the product of the magnitudes of the two vectors times a number that expresses the degree to which $\underline{A}$ and $\underline{B}$ are parallel: $cos\,\theta_{AB}$, where θ_{AB} is the angle between $\underline{A}$ and $\underline{B}$. That is, $\underline{A}.\underline{B} = |\underline{A}||\underline{B}|\,cos\,\theta_{AB}$.



From the previous definition, we must observe that if $\underline{A} \cdot \underline{B} = 0$, this means that the two vectors $\underline{A}$, $\underline{B}$ are perpendicular. Likewise, we must note that

$$\underline{A}.\underline{B} = |\underline{A}|.[projection\ of\ \underline{B}\ in\ the\ \underline{A}\ direction]$$
$$= |\underline{B}|.[projection\ of\ \underline{A}\ in\ the\ \underline{B}\ direction]$$

1.8.1 The properties of the dot product for two vectors

1) $\underline{A} \cdot \underline{B} = \underline{B} \cdot \underline{A}$
2) $\underline{A} \cdot \underline{A} = A^2 = \underline{A}^2$
3) $\underline{i} \cdot \underline{i} = \underline{j} \cdot \underline{j} = \underline{k} \cdot \underline{k} = 1$
4) $\underline{i} \cdot \underline{j} = \underline{j} \cdot \underline{k} = \underline{k} \cdot \underline{i} = 0$
5) $\underline{A} \cdot \underline{B} = (A_x\underline{i} + A_y\underline{j} + A_z\underline{k}) \cdot (B_x\underline{i} + B_y\underline{j} + B_z\underline{k}) = A_xB_x + A_yB_y + A_zB_z$

Also, the angle between the vectors $\underline{A}$, $\underline{B}$ can be determined by

$$cos\,\theta = \frac{\underline{A} \cdot \underline{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{\sqrt{A_x^2 + A_y^2 + A_y^2}\sqrt{B_x^2 + B_y^2 + B_y^2}}$$

Example1.8 Find the unit vector which parallel to the net vector for the two following vectors $\underline{A} = 3\underline{i} - 5\underline{j} + 2\underline{k}, \quad \underline{B} = -\underline{i} + 2\underline{j} + 4\underline{k}$

Solution

Let $\underline{R}$ is a net vector which can be defined as

$$\underline{R} = \underline{A} + \underline{B} = 3\underline{i} - 5\underline{j} + 2\underline{k} - \underline{i} + 2\underline{j} + 4\underline{k} = 2\underline{i} - 3\underline{j} + 6\underline{k}$$

$$\therefore \; R = \sqrt{4 + 9 + 36} = 7$$

Then the required unit vector can be calculated as:

$$\frac{\underline{R}}{R} = \frac{2}{7}\underline{i} - \frac{3}{7}\underline{j} + \frac{6}{7}\underline{k}$$

Example1.9 Find the angle between the two following vectors

$$\underline{A} = 2\underline{i} - \underline{j} + 2\underline{k}, \quad \underline{B} = 3\underline{i} + 4\underline{k}$$

Solution

$$A = \sqrt{4 + 1 + 4} = 3 \quad , \quad B = \sqrt{9 + 16} = 5$$

$$\underline{A} \cdot \underline{B} = (2\underline{i} - \underline{j} + 2\underline{k}) \cdot (3\underline{i} + 4\underline{k}) = 6 + 8 = 14$$

$$\therefore cos\,\theta = \frac{\underline{A} \cdot \underline{B}}{AB} = \frac{14}{15} \Rightarrow \therefore \theta = cos^{-1}\frac{14}{15}$$

Example1.10 Find the angles between $\underline{A} = 4\underline{i} + 6\underline{j} + 7\underline{k}$ and each of the three axes.

Solution

Let the angles between $\underline{A}$ and the *x; y*, and *z* axes be θ, φ and ψ respectively. Then

$$cos\,\theta = \frac{\underline{A}.\underline{i}}{|\underline{A}||\underline{i}|} = \frac{4}{10} = 0.4$$
$$\Rightarrow \theta = cos^{-1}\,0\,.4 = 66.4°$$

Similarly, $\varphi = 53.1°, \psi = 45.6°.$

Example1.11 Prove that $(\underline{A} + \underline{B}).(\underline{A} - \underline{B}) = A^2 - B^2.$

Solution

$$\because \quad (\underline{A} + \underline{B}) \cdot (\underline{A} - \underline{B}) = \underline{A} \cdot \underline{A} - \underline{A} \cdot \underline{B} + \underline{B} \cdot \underline{A} - \underline{B} \cdot \underline{B}$$
$$= A^2 - B^2$$

Example1.12 Find the component of the vector $\underline{F} = 5\underline{i} + 3\underline{j} + 2\underline{k}$ along the vector

$$\underline{r} = 3\underline{i} - 4\underline{j}$$

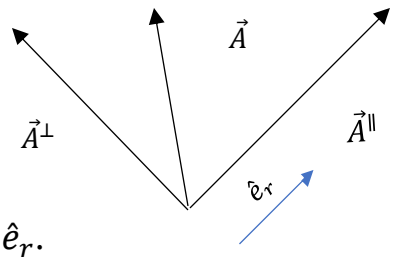
Solution

The dot product of a vector $\vec{F}$ with a unit vector $\hat{e}_r$ gives the projection of the vector $\vec{F}$ in the direction of the unit vector $\hat{e}_r$. Therefore, to find the component of $\vec{F}$ along $\hat{e}_r$, we first find a unit vector $\hat{e}_r$ along $\vec{r}$ and dot it with $\vec{F}$.

$$\Rightarrow \hat{e}_r = \frac{\underline{r}}{|\underline{r}|} = 0.6\underline{i} - 0.8\underline{j}$$
$$\Rightarrow F_r = \vec{F}.\hat{e}_r = 5.4$$

Note that: For any vector $\vec{A}$ and a unit vector $\hat{e}_r$ then the vector $\vec{A}$ can be written as the sum of two parts, $\vec{A}^{\parallel}, \vec{A}^{\perp}$ as

$$\vec{A} = \vec{A}^{\parallel} + \vec{A}^{\perp}$$



where, is parallel $\vec{A}^{\parallel}$ to $\hat{e}_r$ and $\vec{A}^{\perp}$ is perpendicular to $\hat{e}_r$.

$$\vec{A}^{\parallel} = (\vec{A}.\hat{e}_r)\,\hat{e}_r, \qquad \vec{A}^{\perp} = \vec{A} - (\vec{A}.\hat{e}_r)\,\hat{e}_r.$$

1.9 The cross product of two vectors

When you try to move something, you can push and turn it. In mechanics, the measure of your pushing is the net force you apply. The measure of your turning is the net moment, also sometimes called the net torque or net couple. The main mathematical tool here is the vector cross product, a second way of multiplying vectors together. The cross product is used to define (and calculate) moment and to calculate various quantities in dynamics. The cross product also sometimes helps solve three-dimensional geometry problems. So, the cross product is defined as :

$$\therefore \ \underline{A} \ \wedge \ \underline{B} = AB \sin\theta \ \ \underline{n},$$

where $\underline{n}$ can be defined as the unit vector pointing in the direction of the thumb when the fingers are curled from the direction of $\underline{A}$towards the direction of $\underline{B}$.

1.9.1 The properties of the cross product for two vectors

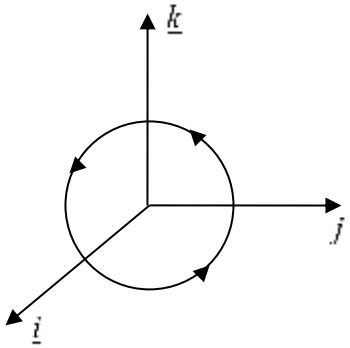
1) $\underline{A} \wedge \underline{B} = -\underline{B} \wedge \underline{A}$

2) $\underline{A} \wedge \underline{A} = 0$

3) $\underline{i} \wedge \underline{i} = \underline{j} \wedge \underline{j} = \underline{k} \wedge \underline{k} = 0$

4) $\underline{i} \wedge \underline{j} = \underline{k} \quad , \quad \underline{j} \wedge \underline{k} = \underline{i} \quad , \quad \underline{k} \wedge \underline{i} = \underline{j}$

$\underline{j} \wedge \underline{i} = -\underline{k} \quad , \quad \underline{k} \wedge \underline{j} = -\underline{i} \quad , \quad \underline{i} \wedge \underline{k} = -\underline{j}$



5) $\underline{A} \wedge \underline{B} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

Theorem. Area of triangles and parallelograms

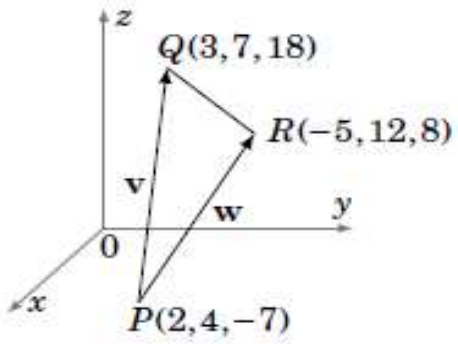
(a) The area *A* of a triangle with adjacent sides **v**, **w** is: $A = |\underline{v} \wedge \underline{w}|/2$.

(b) The area A of a parallelogram with adjacent sides v, w is: $A = |\underline{v} \wedge \underline{w}|$.

Example1.13 Calculate the area of the triangle *PQR*, where *P* = (2,4,−7), *Q* = (3,7,18) and *R* = (−5,12,8).

Solution

Let $v = \underline{PQ}$ and $w = \underline{PR}$, as in the following figure. Then v = (3,7,18)− (2,4,−7) = (1,3,25) and w= (−5,12,8) −(2,4,−7) =(−7,8,15), so the area A of the triangle PQR is $A = |(1,3,25) \wedge (-7,8,15)|/2 = 123.46$

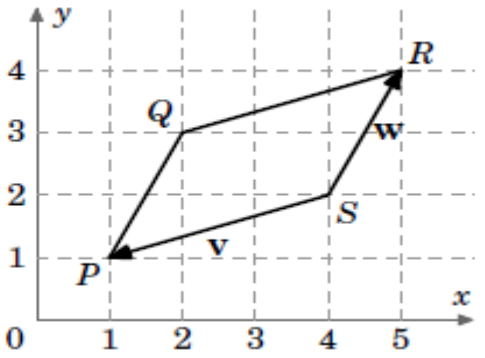


Example1.14 Calculate the area of the parallelogram PQRS, where P = (1,1), Q = (2,3), R = (5,4), and S = (4,2).

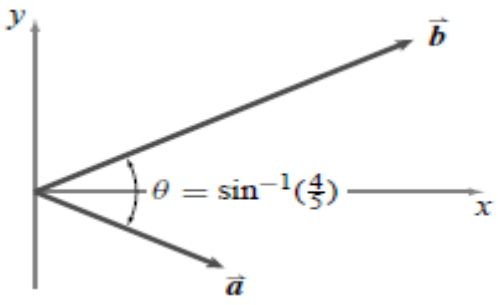
Solution

Let $v = \underline{SP}$ and $w = \underline{SR}$, as in the following figure. Then v = (1,1,0)−(4,2,0) = (-3,-1,0) and w= (5,4,0)−(4,2,0) =(1,2,0), so the area A of the parallelogram PQRS is

$$A = |(-3,-1,0) \wedge (1,2,0)| = 5$$



Example1.15 The two vectors shown in the figure are $\underline{a} = 2\underline{i} - \underline{j}$ and $\underline{b} = 4\underline{i} + 2\underline{j}$. The angle between the two vectors is $\theta = sin^{-1}(4/5)$ (this information can be found out from the given vectors). Find the cross product of the two vectors



(a) Using the angle θ, and

(b) Using the components of the vectors.

Solution

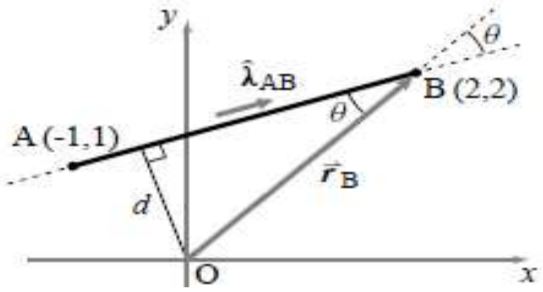
a) Cross product using the angle θ

$$\therefore \ \underline{a} \wedge \underline{b} = |\underline{a}||\underline{b}| \sin\theta \, \underline{n}$$
$$= \sqrt{5}\sqrt{20} \, sin(sin^{-1}(4/5))\underline{k}$$
$$= 8\underline{k}$$

b) Cross product using the components of the vectors.

$$\therefore\ \underline{a} \wedge \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -1 & 0 \\ 4 & 2 & 0 \end{vmatrix} = 8\underline{k}$$

Example1.16 A straight line passes through two points, A (-1,4) and B (2,2), in the xy plane. Find the shortest distance from the origin to the line.



Solution

$$\because\ \underline{r} \wedge \underline{AB} = |\underline{r}||\underline{AB}|\, sin\,\theta\, \underline{k}$$
$$= d|\underline{AB}|\underline{k}$$
$$\Rightarrow d\underline{k} = \underline{r} \wedge \frac{\underline{AB}}{|\underline{AB}|} = \underline{r} \wedge \underline{e}_{AB}$$
$$\Rightarrow |d\underline{k}| = d = \left|\underline{r} \wedge \frac{\underline{AB}}{|\underline{AB}|}\right| = |\underline{r} \wedge \underline{e}_{AB}|$$
$$= \frac{1}{\sqrt{10}}|4\underline{k}| = \frac{4}{\sqrt{10}}$$
$$\Rightarrow d = \frac{4}{\sqrt{10}}$$

Example1.16 Prove that $(\underline{A} \cdot \underline{B})^2 = A^2B^2 - |\underline{A} \wedge \underline{B}|^2$

Solution

$$\therefore \underline{A} \cdot \underline{B} = A\ B\ cos\,\theta \quad (1)$$
$$\underline{A} \wedge \underline{B} = A\ B\ sin\,\theta\ \ \underline{n}$$
$$\therefore |\underline{A} \wedge \underline{B}| = A\ B\ sin\,\theta \quad (2)$$

By squaring Eqs. (1) and (2) and then adding the results we have:

$$(\underline{A} \cdot \underline{B})^2 + |\underline{A} \wedge \underline{B}|^2 = A^2B^2(cos^2\,\theta + sin^2\,\theta) = A^2B^2$$
$$\therefore (\underline{A} \cdot \underline{B})^2 = A^2B^2 - |\underline{A} \wedge \underline{B}|^2$$

Example1.17 Find the unit vector which perpendicular on both two following vectors

$$\underline{A} = \underline{j} + \underline{k}, \quad \underline{B} = 6\underline{i} - 3\underline{j} + 4\underline{k}$$

Solution

$$\underline{A} \wedge \underline{B} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 1 & 1 \\ 6 & -3 & 4 \end{vmatrix} = 7\underline{i} + 6\underline{j} - 6\underline{k}$$

$$\therefore \ |\underline{A} \wedge \underline{B}| = \sqrt{49 + 36 + 36} = 11$$

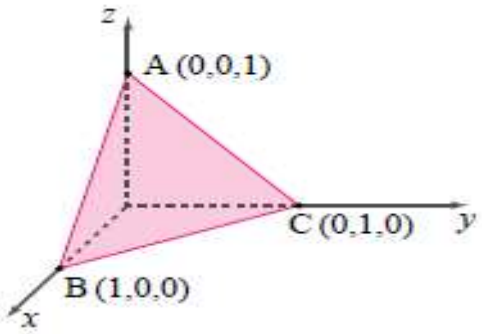
$$\Rightarrow \underline{e} = \frac{\underline{A} \wedge \underline{B}}{|\underline{A} \wedge \underline{B}|} = \frac{7}{11}\underline{i} + \frac{6}{11}\underline{j} - \frac{6}{11}\underline{k}$$

Example1.18 Prove that $(\underline{A} + \underline{B}) \wedge (\underline{A} - \underline{B}) = 2\underline{B} \wedge \underline{A}$

Solution

$$\begin{aligned}(\underline{A} + \underline{B}) \wedge (\underline{A} - \underline{B}) &= \underline{A} \wedge \underline{A} - \underline{A} \wedge \underline{B} + \underline{B} \wedge \underline{A} - \underline{B} \wedge \underline{B} \\ &= 0 + \underline{B} \wedge \underline{A} + \underline{B} \wedge \underline{A} + 0 \\ &= 2\underline{B} \wedge \underline{A}\end{aligned}$$

Example1.19 Find a unit vector normal to the plane ABC shown in the figure.



Solution

A vector normal to the plane ABC would be normal to any vector in that plane. In particular, if we take any two vectors, say $\vec{r}_{AB}$ and $\vec{r}_{AC}$, the normal to the plane would be perpendicular to both $\vec{r}_{AB}$ and $\vec{r}_{AC}$. Since the cross product of two vectors gives a vector perpendicular to both vectors, we can find the desired normal vector by taking the cross product of $\vec{r}_{AB}$ and $\vec{r}_{AC}$. Thus,

$$\underline{N} = \vec{r}_{AB} \wedge \vec{r}_{AC} = \underline{i} + \underline{j} + \underline{k}$$

$$\Rightarrow \underline{n} = \frac{\underline{N}}{|\underline{N}|} = \frac{1}{\sqrt{3}}(\underline{i} + \underline{j} + \underline{k})$$

1.9.2 The Triple Scalar Product

The triple scalar product for any three vectors $\underline{A}, \underline{B}, \underline{C}$ can be defied as $\underline{A} \cdot (\underline{B} \wedge \underline{C})$

$$\because \underline{B} \wedge \underline{C} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

$$\Rightarrow \underline{A} \cdot (\underline{B} \wedge \underline{C}) = (A_x\underline{i} + A_y\underline{j} + A_z\underline{k}) \cdot \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

$$= A_x(B_yC_z - B_zC_y) + A_y(B_zC_x - B_xC_z) + A_z(B_xC_y - B_yC_x)$$

$$= \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

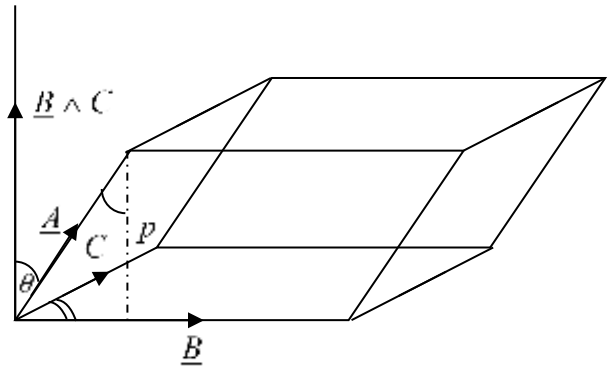
1.9.3 The Properties of the Triple Scalar Product

$$\underline{A} \cdot (\underline{B} \wedge \underline{C}) = (\underline{B} \wedge \underline{C}) \cdot \underline{A} = \underline{B} \cdot (\underline{C} \wedge \underline{A}) = (\underline{C} \wedge \underline{C}) \cdot \underline{B} = \underline{C} \cdot (\underline{A} \wedge \underline{B}) = (\underline{A} \wedge \underline{B}) \cdot \underline{C}$$

Also,

$$\underline{A} \cdot (\underline{A} \wedge \underline{C}) = (\underline{A} \wedge \underline{A}) \cdot \underline{C} = 0$$

Result. The triple scalar product for any three sides $\underline{A}, \underline{B}, \underline{C}$ of a parallelepiped is equal to its volume.



1.10 The Triple Vector Product

The triple vector product for any three vectors $\underline{A}, \underline{B}, \underline{C}$ can be defied as $\underline{A} \wedge (\underline{B} \wedge \underline{C})$.

Theorem. For any three vectors $\underline{A}, \underline{B}, \underline{C}$ we have:

$$\underline{A} \wedge (\underline{B} \wedge \underline{C}) = (\underline{A} \cdot \underline{C})\underline{B} - (\underline{A} \cdot \underline{B})\underline{C}$$

Proof:

$$\because \ \underline{B} \wedge \underline{C} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

$$\therefore \underline{A} \wedge (\underline{B} \wedge \underline{C}) =$$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ A_x & A_y & A_z \\ B_yC_z - B_zC_y & B_zC_x - B_xC_z & B_xC_y - B_yC_x \end{vmatrix}$$

$$= \underline{i}\left[A_y(B_xC_y - B_yC_x) - A_z(B_zC_x - B_xC_z)\right]$$
$$+\underline{j}\left[A_z(B_yC_z - B_zC_y) - A_x(B_xC_y - B_yC_x)\right]$$
$$+\underline{k}\left[A_y(B_yC_z - B_zC_y) - A_x(B_zC_x - B_xC_z)\right]$$

$$= (\underline{A} \cdot \underline{C})\underline{B} - (\underline{A} \cdot \underline{B})\underline{C}$$

Example1.20 Prove that $\underline{A} \wedge (\underline{B} \wedge \underline{C}) + \underline{B} \wedge (\underline{C} \wedge \underline{A}) + \underline{C} \wedge (\underline{A} \wedge \underline{B}) = 0$

Solution

$$L.H.S = (\underline{A} \cdot \underline{C})\underline{B} - (\underline{A} \cdot \underline{B})\underline{C}$$
$$+ (\underline{B} \cdot \underline{A})\underline{C} - (\underline{B} \cdot \underline{C})\underline{A}$$
$$+ (\underline{C} \cdot \underline{B})\underline{A} - (\underline{C} \cdot \underline{A})\underline{B} = 0$$

Exercises

1. Calculate the magnitudes of the following vectors:
   (a) $\mathbf{v} = (2,-1)$ (b) $\mathbf{v} = (2,-1,0)$ (c) $\mathbf{v} = (3,2,-2)$ (d) $\mathbf{v} = (0,0,1)$ (e) $\mathbf{v} = (6,4,-4)$

2. For the points $P = (1,-1,1)$, $Q = (2,-2,2)$, $R = (2,0,1)$, $S = (3,-1,2)$, does $\overrightarrow{PQ} = \overrightarrow{RS}$?

3. For the points $P = (0,0,0)$, $Q = (1,3,2)$, $R = (1,0,1)$, $S = (2,3,4)$, does $\overrightarrow{PQ} = \overrightarrow{RS}$?

4. If $ABCD$is a parallelogram, prove that $\underline{AC} + \underline{DB} = 2\underline{AB}$.

5. If $ABCDEF$is a regular hexagonal with a center G prove that
$$\underline{AB} + \underline{AC} + \underline{AE} + \underline{AF} = 4\ \underline{AG}.$$
Also, for any point o prove that $\underline{oA} + \underline{oB} + \underline{oC} + \underline{oD} + \underline{oE} + \underline{oF} = 6\ \underline{oG}$.

6. Find the value h which makes the following two vectors to be orthogonal
$$\underline{A} = 3\underline{i} + \underline{j} - \underline{k} \quad , \quad \underline{B} = h\underline{i} - 4\underline{j} + 8\underline{k}.$$

7. Find the angle between the following two vector
$$\underline{A} = 3\underline{i} - 2\underline{j} + \underline{k} \quad , \quad \underline{B} = 2\underline{i} + \underline{j} + 3\underline{k}.$$

8. If $\underline{A} = \underline{i} + 2\underline{j} - \underline{k}$, $\underline{B} = 2\underline{i} - \underline{j} + 3\underline{k}$, $\underline{C} = 3\underline{i} + \underline{j} + 2\underline{k}$, then calculate

$$\underline{A} \cdot \underline{B} \quad , \quad \underline{A} \wedge \underline{B} \quad , \quad \underline{A} \cdot (\underline{B} \wedge \underline{C}) \quad , \quad \underline{A} \wedge (\underline{B} \wedge \underline{C})$$

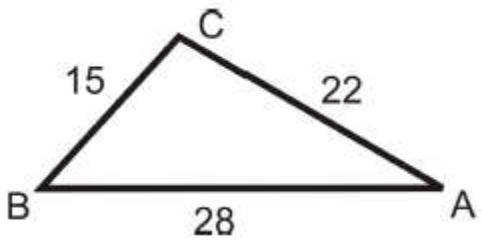
9. If $\underline{P} \cdot \underline{Q} = 0$, $\underline{P} \cdot \underline{R} = 0$, prove that $\underline{P} \wedge (\underline{Q} \wedge \underline{R}) = 0$.

10. Prove that $\left[\underline{A} \wedge (\underline{B} \wedge \underline{C})\right] \wedge \underline{C} = (\underline{A} \cdot \underline{C})(\underline{B} \wedge \underline{C})$.

11. Prove that $\underline{D} \cdot \left[\underline{A} \wedge \left\{\underline{B} \wedge (\underline{C} \wedge \underline{D})\right\}\right] = (\underline{B} \cdot \underline{D})\left[(\underline{C} \wedge \underline{D}) \cdot \underline{A}\right]$.

12. Calculate the volume of the parallelepiped which is represented by the following three sides $\underline{A} = 2\underline{i} - \underline{j} + 2\underline{k}$, $\underline{B} = \underline{i} - \underline{j} + 2\underline{k}$, $\underline{C} = 3\underline{i} + 2\underline{j} - \underline{k}$.

13. Find the following angles in the triangle below.



Chapter 2
Forces & Moments

A force is defined as an action of one body on another. In dynamics we will see that a force is defined as an action which tends to cause acceleration of a body. A force is a vector quantity, because its effect depends on the direction as well as on the magnitude of the action.

2.1 Rectangular Components

The most common two-dimensional resolution of a force vector is into rectangular components. It follows from the parallelogram rule

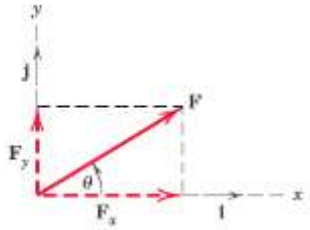


Fig. 2.1

that the vector $\vec{F}$ of Fig. 2.1 may be written as

$$\vec{F} = \vec{F}_x + \vec{F}_y$$

where $\vec{F}_x$ and $\vec{F}_y$ are vector components of F in the *x*- and *y*-directions. In terms of the unit vectors $\hat{\imath}$ and $\hat{\jmath}$, and thus we may write

$$\vec{F} = F_x\,\hat{\imath} + F_y\,\hat{\jmath}$$

where

$$F_x = F\cos\theta \qquad F = \sqrt{F_x^{\,2} + F_y^{\,2}}$$

$$F_y = F\sin\theta \qquad \theta = \tan^{-1}\frac{F_y}{F_x}$$

A particle is said to be in equilibrium under effect of forces $\overrightarrow{F_1}, \overrightarrow{F_2}, \ldots, \overrightarrow{F_n}$ if

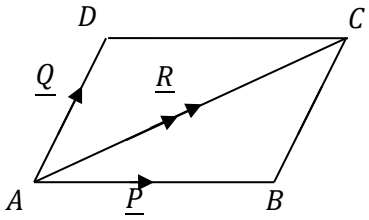
$$\sum F = \sum F_x\hat{\imath} + \sum F_y\hat{\jmath} = 0$$

The two scalar equations of force equilibrium can be applied with reference to an established *x*, *y* coordinate system

$$\sum F_x = \sum F_y = 0$$

2.2 Resultant force

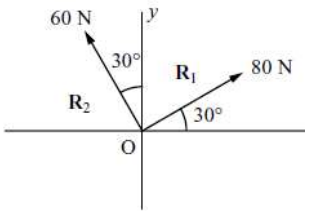
The resultant force is the single force that represents a system of forces. Forces can act in any direction, all affecting the acceleration of an object. So, if $\underline{P}$, $\underline{Q}$ are two



forces acting on a point A with a direction $AB,\ AD$; respectively. Then the resultant force can be represented on the line AC and acts on the same point A (see the following figure).

Example 2.1 Find the resultant of the forces R_1 and R_2.

Solution

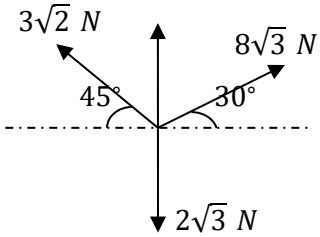


$$\mathbf{R} = \sqrt{R_1^2 + R_2^2}$$

$$\mathbf{R} = 10\sqrt{64+36} = 100 \text{ N}$$

Example 2.2 Find the resultant of the forces which appear in the following diagram.

Solution



$$X = 8\sqrt{3}\,cos\,30° - 3\sqrt{2}\,cos\,45°$$
$$= 8\sqrt{3} \times \frac{\sqrt{3}}{2} - 3\sqrt{2} \times \frac{1}{\sqrt{2}}$$
$$= 12 - 3 = 9 \ \ N$$

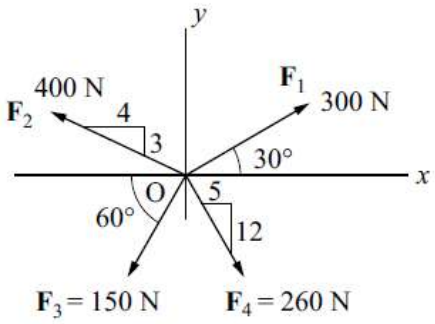
Also,
$$Y = 8\sqrt{3}\,sin\,30° + 3\sqrt{2}\,sin\,45° - 2\sqrt{3}$$
$$= 8\sqrt{3} \times \frac{1}{2} + 3\sqrt{2} \times \frac{1}{\sqrt{2}} - 2\sqrt{3}$$
$$= (2\sqrt{3} + 3)\ N$$

then
$$R = \sqrt{X^2 + Y^2} = \sqrt{9^2 + (2\sqrt{3}+3)^2} = 11.1\ N$$

and
$$tan\,\theta = Y/X = (2\sqrt{3}+3)/9 \quad \Rightarrow \quad \therefore \quad \theta = 35.69°$$

Example 2.3 Determine the resultant of four forces acting on a body as shown below.

Solution



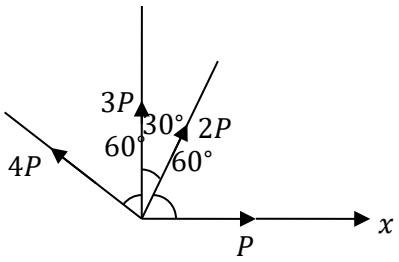
$$\mathrm{R}^x = 259.8 - 320 - 75 + 100 = -35.2 \text{ N}$$

$$\mathrm{R}^y = 150 + 240 - 130 - 240 = 20 \text{ N}$$

$$\mathbf{R} = \sqrt{(-35.2)^2 + (20)^2}$$

$$\tan\theta = \frac{R^y}{R^x} = \frac{20}{35.2}$$
$$\theta = 29.6°$$

Example 2.4 Determine the magnitude and the direction of the resultant force for four forces $4P, 3P, 2P, P$ acting on a body, where the angle between the two forces $2P, P$ is $60°$, the angle between the two forces $3P, 2P$ is $30°$and the angle between the two forces $4P, 3P$ is $60°$.



Solution

Let the body exists on the origin o of the xy-coordinates, so we observe that from the following diagram

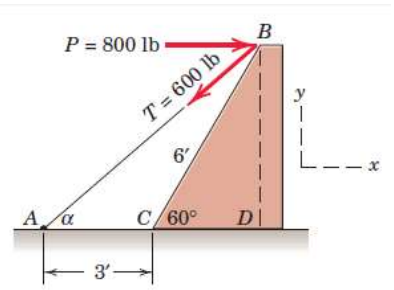
$$X = P + 2P\,cos\,6\,0° - 4P\,cos\,3\,0°$$
$$= P(1 + 1 - 2\sqrt{3}) = -1.46P \quad N$$

also, $$Y = 3P + 2P\,sin\,6\,0° + 4P\,sin\,3\,0°$$
$$= P(3 + \sqrt{3} + 2) = 6.73P \quad N$$

then $$\therefore \; R = \sqrt{X^2 + Y^2} = 6.9P \; N$$

and $$tan\,\theta = \frac{Y}{X} = -4.6 \quad \Rightarrow \quad \therefore \; \theta = 102.27°$$

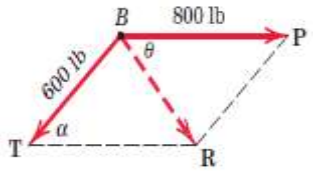
Example 2.5 Combine the two forces P and T, which act on the fixed structure at B, into a single equivalent force R.



Solution

$$\tan\alpha = \frac{\overline{BD}}{\overline{AD}} = \frac{6\sin 60^\circ}{3 + 6\cos 60^\circ} = 0.866 \qquad \alpha = 40.9^\circ$$

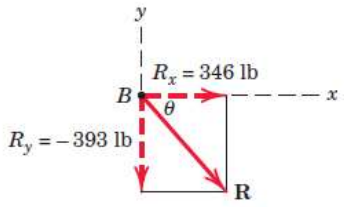
By using the *x-y* coordinate system on the given figure, we may write



$$R_x = 800 - 600\,cos\,40.9^\circ = 346\ lb$$

$$R_y = -600\,sin\,40.9^\circ = -393\ lb$$

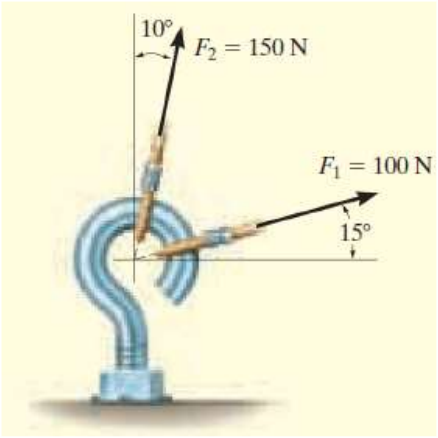
The magnitude and direction of the resultant force *R* are:



$$R = \sqrt{(346)^2 + (-393)^2} = 524\ lb$$

$$\theta = \tan^{-1}\frac{393}{346} = 48.6^\circ$$

Example 2.6 The screw eye in opposite Fig. is subjected to two forces, $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$, determine the magnitude and direction of the resultant force.



Solution

$$\overrightarrow{F_1} = 100\ N, \theta_1 = 15^\circ, \qquad \overrightarrow{F_2} = 150\ N, \theta_2 = 10^\circ$$

$$R_x = 100\cos 15^\circ + 150\,sin\ 10^\circ = 122.63\ lb$$

$$R_y = 100\sin 15^\circ + 150\,cos\ 10^\circ = 173.6\ lb$$

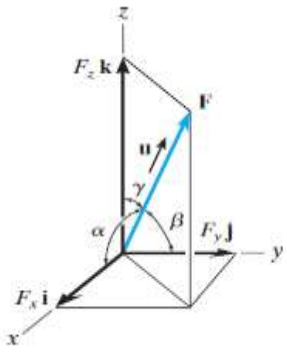
The magnitude and direction of the resultant force *R* are:

$$R = \sqrt{(122.63)^2 + (173.6)^2} = 212.54\ lb$$

$$\theta = \tan^{-1}\frac{173.6}{122.6} = 54.76^\circ$$

2.3 Three-Dimensional Force Systems

In three dimensions, a force can be resolved into its Cartesian components along the *x*, *y*, *z* axes so that



$$\vec{F} = F_x\,\hat{\imath} + F_y\,\hat{\jmath} + F_z\,\hat{k}$$

The magnitude of $\vec{F}$ is determined as

$$\left|\vec{F}\right| = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

The coordinate direction angles α, β, γ are determined by formulating a unit vector in the direction of F.

$$\cos\alpha = \frac{F_x}{|\vec{F}|}, \cos\beta = \frac{F_y}{|\vec{F}|}, \cos\gamma = \frac{F_z}{|\vec{F}|}$$

The coordinate direction angles are related so that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

the resultant of a concurrent force system

$$\mathbf{F}_R = \sum \boldsymbol{F}_x + \sum \boldsymbol{F}_y + \sum \boldsymbol{F}_z$$

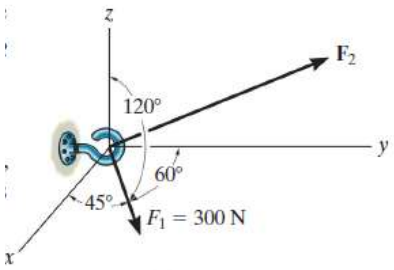
A particle is said to be in equilibrium if it remains at rest if originally at rest or has a constant velocity if originally in motion. This condition is stated by the equation of equilibrium, $\sum F = 0$, where $\sum F$ is the vector sum of all the forces acting on the particle. In the case of a three-dimensional force system,

$$\sum F = \sum F_x\hat{\imath} + \sum F_y\hat{\jmath} + \sum F_z\,\hat{k} = 0$$

this equation we require

$$\sum F_x = \sum F_y = \sum F_z = 0$$

Example 2.7 Two forces act on the hook shown in the opposite Fig. Specify the magnitude of $\mathbf{F}_2$ and its coordinate direction angles so that the resultant force $\mathbf{F}_R$ acts along the positive y axis and has a magnitude of 800 N.



Solution

Let the resultant force $\mathbf{F}_R = \mathbf{F}_1 + \mathbf{F}_2$, where

$$\begin{aligned}\boldsymbol{F}_1 &= 300cos\ 45°\boldsymbol{i} + 300cos\ 60°\boldsymbol{j} + 300cos\ 120°\boldsymbol{k}\\ &= \{212.1\boldsymbol{i} + 150\boldsymbol{j} - 150\boldsymbol{k}\}N\\ \boldsymbol{F}_2 &= F_{2x}\boldsymbol{i} + F_{2y}\boldsymbol{j} + F_{2z}\boldsymbol{k}\end{aligned}$$

Since $\mathbf{F}_R$ has a magnitude of 800 N and acts in the $+\boldsymbol{j}$ direction,

$$\mathbf{F}_R = (800\,\mathrm{j}\,)\mathrm{N}$$

We require

$$\begin{aligned} \because \mathbf{F}_R &= \mathbf{F}_1 + \mathbf{F}_2 \\ \therefore 800\mathbf{j} &= 212.1\mathbf{i} + 150\mathbf{j} - 150\mathbf{k} + F_{2x}\mathbf{i} + F_{2y}\mathbf{j} + F_{2z}\mathbf{k} \end{aligned}$$

$$800\mathbf{j} = (212.1 + F_{2x})\mathbf{i} + \left(150 + F_{2y}\right)\mathbf{j} + (-150 + F_{2z})\mathbf{k}$$

$$\begin{aligned} 0 &= 212.1 + F_{2x} \quad & F_{2x} &= -212.1 \text{ N} \\ 800 &= 150 + F_{2y} & F_{2y} &= 650 \text{ N} \\ 0 &= -150 + F_{2z} & F_{2z} &= 150 \text{ N} \end{aligned}$$

The magnitude of $\mathbf{F}_2$ is thus

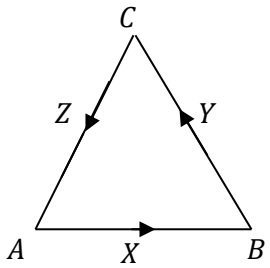
$$\begin{aligned} F_2 &= \sqrt{(-212.1 \text{ N})^2 + (650 \text{ N})^2 + (150 \text{ N})^2} \\ &= 700 \text{ N} \end{aligned}$$

To determine the direction of F_2, we introduce the angles $\alpha_2, \beta_2, \gamma_2$

$$\begin{aligned} \cos\,\alpha_2 &= \frac{-212.1}{700}; \quad & \alpha_2 &= 108^\circ \\ \cos\,\beta_2 &= \frac{650}{700}; & \beta_2 &= 21.8^\circ \\ \cos\,\gamma_2 &= \frac{150}{700}; & \gamma_2 &= 77.6^\circ \end{aligned}$$

2.4 Equilibrium Under The Effect of Three Forces

Assume that X, Y, Z are three forces which acts on a particle in an equilibrium state. These forces can be represented by the sides of the following triangle ABC. So, we observe that:



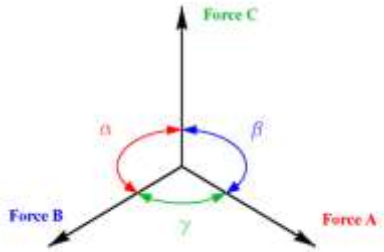
$$\frac{AB}{\sin \hat{BCA}} = \frac{BC}{\sin \hat{CAB}} = \frac{CA}{\sin \hat{ABC}}$$

$$\begin{aligned} \hat{BCA} &= 180^\circ - \alpha \\ \hat{CAB} &= 180^\circ - \beta \\ \hat{ABC} &= 180^\circ - \gamma \end{aligned}$$

$$\frac{AB}{sin(\,180^\circ - \alpha)} = \frac{BC}{sin(\,180^\circ - \beta)} = \frac{CA}{sin(\,180^\circ - \gamma)}$$

$$\frac{AB}{\sin \alpha} = \frac{BC}{\sin \beta} = \frac{CA}{\sin \gamma}$$

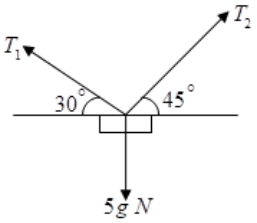
$$\frac{X}{\sin\alpha} = \frac{Y}{\sin\beta} = \frac{Z}{\sin\gamma}$$



2.4.1 Lami's Theorem

In static, Lami's theorem is an equation that relates the magnitudes of three coplanar, concurrent and non-collinear forces, that keeps a body in static equilibrium. Lami's theorem states that if three forces acting at a point are in equilibrium, each force is proportional to the sine of the angle between the other two forces. Consider three forces A, B, C acting on a particle or rigid body making angles α, β and γ with each other.

Example 2.8 A mass with magnitude which hanging down by two inflexible strings. If these strings make the angles with the horizontal axis. Find the tensions in each string.



Solution

We observe that the mass is in an equilibrium state under three forces. So, after applying Lami's law to get the pull forces T_1, T_2 in two strings.

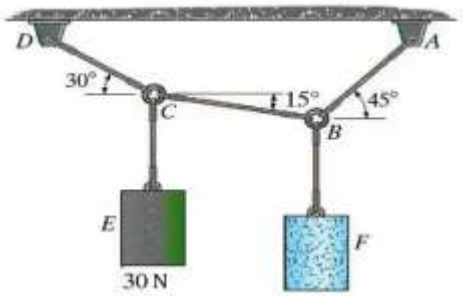
$$\frac{T_1}{\sin(90^\circ + 45^\circ)} = \frac{T_2}{\sin(90^\circ + 30^\circ)} = \frac{5g}{\sin(180^\circ - 30^\circ - 45^\circ)}$$

$$\frac{T_1}{\sin 135^\circ} = \frac{T_2}{\sin 120^\circ} = \frac{5g}{\sin 105^\circ}$$

$$\therefore \quad T_1 = \frac{5g \sin 135^\circ}{\sin 105^\circ} = 35.87\ N$$

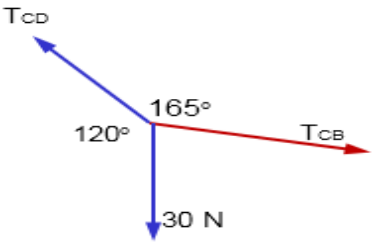
$$T_2 = \frac{5g \sin 120^\circ}{\sin 105^\circ} = 43.93\ N$$

Example 2.9 Two loads of 30 N and W_F are hung by a set of wire ropes as shown in figure. Determine the tensions in the wire ropes AB, BC and CD. Also determine the unknown weight W_F hung at the point B.

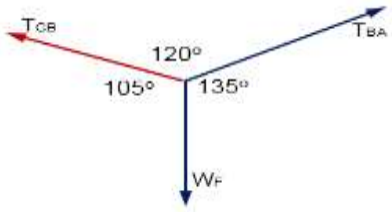


Solution

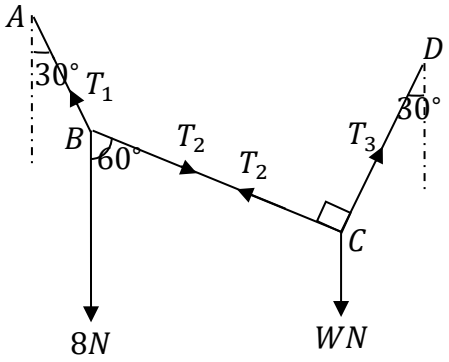
All the included angles between the wire ropes CD, CE and CB are displayed in the free body diagram as shown in figure. Now considering the equilibrium of the portion CD, CE and CB, and applying Lami's theorem, we get the tension in the wire rope CB, T_{CB} = 100.38 N and the tension in the string BC is TCD = 111.96 N.



Now the equilibrium of the second portion CB, BF and BA is considered, by applying Lami's theorem, we get the tension in the wire rope CB, T_{BA} = 137.12N and the unknown weight W_F = 122.94 N



Example 2.10 If $ABCD$ is a string, the mass $8(kg.wt = N)$ hanging down at B and the mass $W(kg.wt = N)$ is hanging down. If AB, BC, CD are parts of the string which make the angles $30°, 60°, 30°$ with the vertical axis. Find W and the tensions in each part of the string.



Solution

Suppose that T_1, T_2, T_3 are the pulling force in AB, BC, CD; respectively. Suppose the equilibrium at $B$and by applying Lami's law, then we have:

$$\frac{T_1}{sin\,60°} = \frac{T_2}{sin\,150°} = \frac{8}{sin\,150°}$$

$$T_1 = \frac{8\,sin\,60°}{sin\,150°} = \frac{8(\sqrt{3}/2)}{1/2} = 8\sqrt{3}\ N$$

$$T_2 = \frac{8\,sin\,150°}{sin\,150°} = 8\ N$$

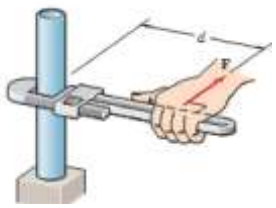
Also, by assume the equilibrium at C for the load W and by applying Lami's law, then we have:

$$\frac{T_2}{sin\,150°} = \frac{T_3}{sin\,120°} = \frac{W}{sin\,90°}$$

$$\therefore\ T_3 = 8\sqrt{3}\ N \quad , \quad W = 16\ N$$

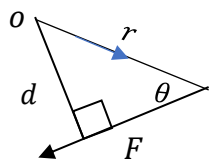
2.5 Moment of Force

When a force is applied to a body it will produce a tendency for the body to rotate about a point that is not on the line of action of the force. This tendency to rotate is sometimes called a ***torque***, but most often it is called the moment of a force or simply the ***moment***. So, the moment of a force $\underline{F}$ about a point o can be defined as



$$\underline{M_o} = \underline{r} \wedge \underline{F},$$

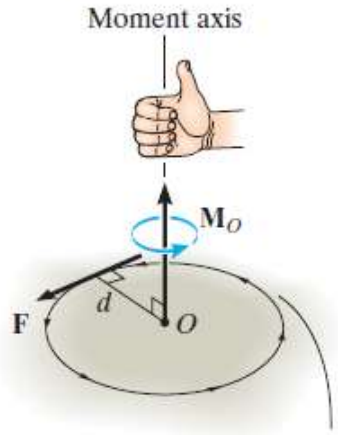
where, $\underline{r}$ is a position vector relative to a point "o" for any point on the force.



Also, the magnitude for the moment is equal to the following relation

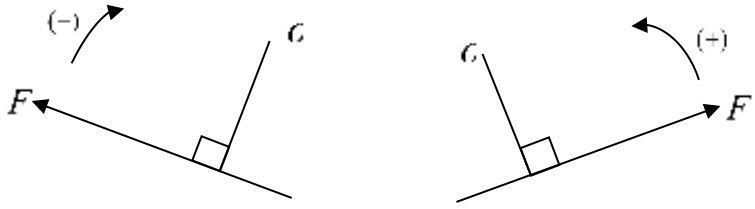
$$\left|\underline{M_o}\right| = M_o = Fr\sin\theta = Fd,$$

where d is the moment arm or perpendicular distance from the axis at point "o" to the line of action of the force. Units of moment is *N.m* or *lb.ft*. The direction of $M_\circ$ is defined by its moment axis, which is perpendicular to the plane that contains the force F and its moment arm d. The right-hand rule is used to establish the sense of direction of M_o.

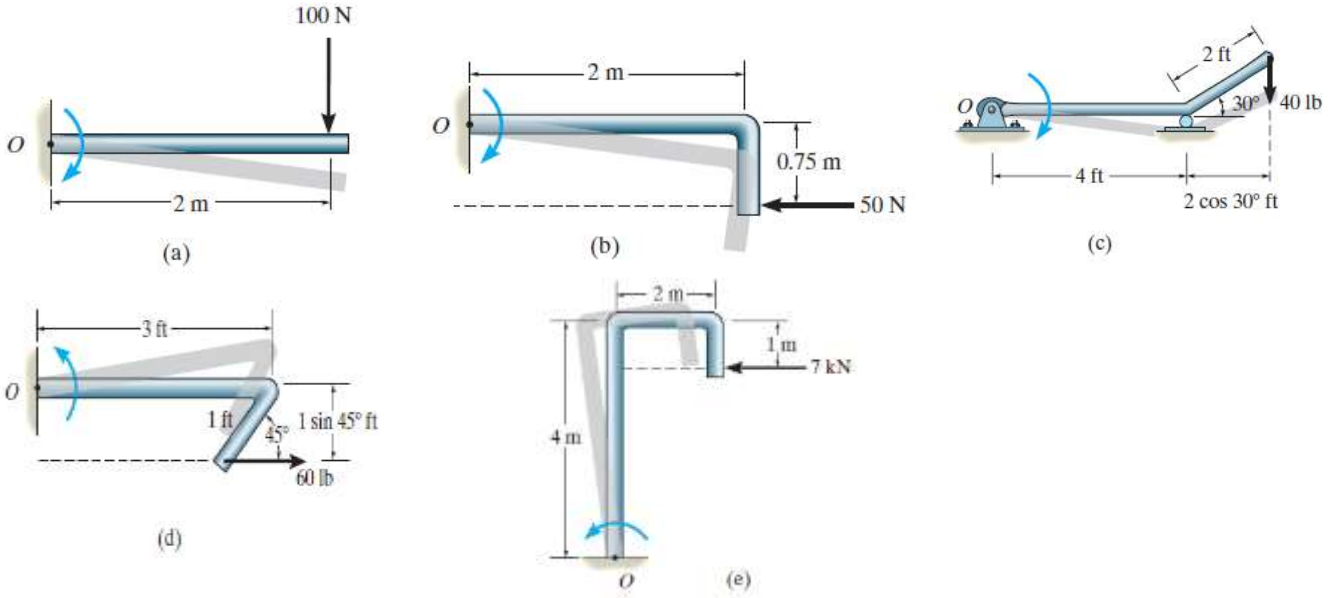


Note that: Closing the Door (Moment = Force x Perpendicular Distance

When dealing with moments it is important to apply a consistent sign convention with the direction of the moment. Look at which way the moment is turning. Is it turning *clockwise* or *anticlockwise?*. For the purposes of this activity, let us apply a sign convention: if a moment is moving *anticlockwise* it is considered a *positive* moment. But if a moment is moving *clockwise* it is considered a *negative* moment.



Example 2.11 For each case determine the moment about the point "o"



Solution

(a) $M_o = (100N)(2m) = 200\ N.m$

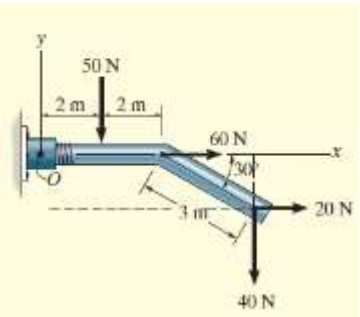
(b) $M_o = (50N)(0.75\ m) = 37.5\ N.m$

(c) $M_o = (40\ lb)(4ft + 2\cos 30°\ ft) = 229\ lb.ft$

(d) $M_o = (60\ lb)(1\sin 45°\ ft) = 42.4\ lb.ft$

(e) $M_o = (7\ kN)(3\ m) = 21\ kN.m$

Example 2.12 Determine the moment about the point "o" for the opposite Fig.

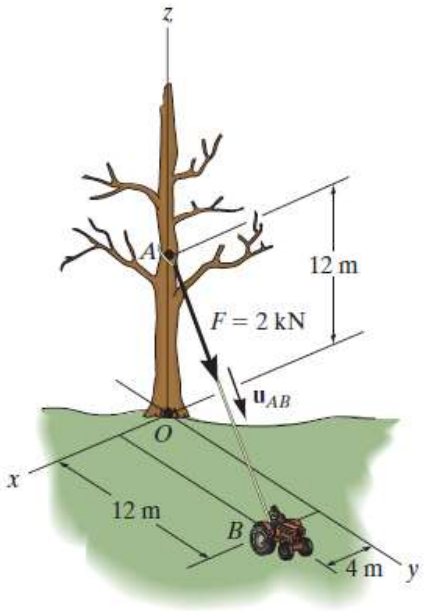


Solution

Assuming that positive moments act in the $+\hat{k}$ direction, i.e., counterclockwise, we have

$$\begin{aligned} M_o &= \Sigma Fd \\ &= -50\ \text{N}(2\ \text{m}) + 60\ \text{N}(0) + 20\ \text{N}(3\sin\ 30°\text{m}) \\ &\quad -40\ \text{N}(4\ \text{m} + 3\cos\ 30°\text{m}) \\ &= -334\ \text{N}\cdot\text{m} = 334\ \text{N}\cdot\text{m} \end{aligned}$$

Example 2.13 Determine the moment produced by the force F in the opposite about point *O*.



Solution

The position vectors $\mathbf{r}_A$ and $\mathbf{r}_B$ are

$$\overrightarrow{\mathbf{r}_A} = 12\,\hat{k} \quad \text{and} \quad \overrightarrow{\mathbf{r}_B} = 4\,\hat{\imath} + 12\,\hat{\jmath}$$

Force F expressed as a Cartesian vector is

$$\vec{\mathrm{F}} = F\mathrm{r}_{AB} = 2\left[\frac{4\mathrm{i} + 12\mathrm{j} - 12\mathrm{k}}{\sqrt{16 + 144 + 144}}\right]$$

$$= (0.4588\,\mathrm{i} + 1.376\,\mathrm{j} - 1.376\,\mathrm{k})kN$$

$$\because \mathrm{M}_O = \mathrm{r}_A \times \mathrm{F} = \begin{vmatrix} \mathrm{i} & \mathrm{j} & \mathrm{k} \\ 0 & 0 & 12 \\ 0.4588 & 1.376 & -1.376 \end{vmatrix}$$

$$= (-16.5\,\mathrm{i} + 5.51\,\mathrm{j})kN \cdot m$$

2.6 Moment of a Force about a Specified Axis

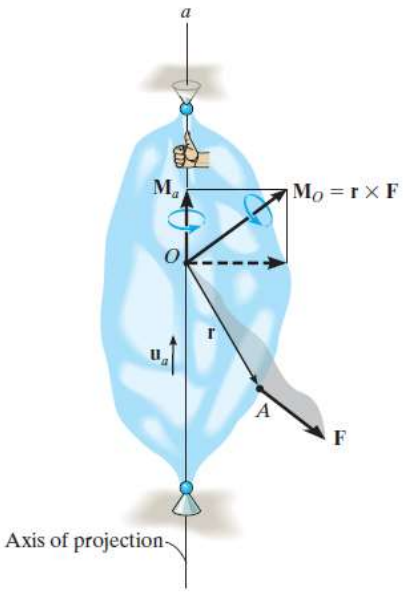
Sometimes, the moment produced by a force about a *specified axis* must be determined. Let a force F acts on a point A. Then, the moment of force F about O

$$\underline{M_o} = \underline{r} \wedge \underline{F},$$

and the projection of this moment M_o onto a axis is

$$\underline{M_a} = u_a.(\underline{r} \wedge \underline{F}),$$

where, u_a is the unit vector that specifies the direction of the *a* axis



$$M_a = \mathbf{u}_a \cdot (\mathbf{r} \times \mathbf{F}) = \begin{vmatrix} u_{a_x} & u_{a_y} & u_{a_z} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

Example 2.14 Determine the moment of the force $F = 2i + 3j - k$ acting through the point (3,1,1) with respect to the line passing from $(2,5,-2)$ through $(3,-1,1)$. The coordinates are in feet.

Solution

The moment arm r may be found by using the vector from-either point on the line to the point on the force. From $(2,5,-2)$ the vector $\mathrm{r} = \mathrm{i} - 4\mathrm{j} + 3\mathrm{k}$. The moment M about the chosen point is

$$M = r \times F = \begin{vmatrix} i & j & k \\ 1 & -4 & 3 \\ 2 & 3 & -1 \end{vmatrix} = -5i + 7j + 11k$$

the unit vector for the line passing from $(2,5,-2)$ through $(3,-1,1)$.

$$\hat{e} = \frac{i - 6j + 3k}{\sqrt{1^2 + 6^2 + 3^2}} = \frac{i - 6j + 3k}{\sqrt{46}}$$

hence, the moment of the force F about the line

$$M_l = M.\hat{e} = (-5i + 7j + 11k).\left(\frac{i - 6j + 3k}{\sqrt{46}}\right) = -2.06\ lf.ft$$

Example 2.15 Find the moment of the force $20\ N$which effect on the straight line $\overrightarrow{AB}$, where $A(5,3,-3), B(2,2,1)$ about the origin.

Solution

Firstly, we must get a unit vector $\hat{e}$ in the direction of $\overrightarrow{AB}$ as follows:

$$\hat{e} = \frac{\overrightarrow{AB}}{\left|\overrightarrow{AB}\right|} = \frac{B - A}{\left|\overrightarrow{AB}\right|} = \frac{(2-5)\underline{i} + (2-3)\underline{j} + (-1+3)\underline{k}}{\sqrt{(2-5)^2 + (2-3)^2 + (-1+3)^2}}$$

$$= \frac{1}{\sqrt{14}}(-3\underline{i} - \underline{j} + 2\underline{k})$$

So, $$\vec{F} = \frac{20}{\sqrt{14}}(-3\underline{i} - \underline{j} + 2\underline{k})$$

Suppose that $\vec{r}$ is a position vector for the point B, then

$$\vec{r} = 2\hat{\imath} + 2\,\hat{\jmath} - \hat{k}$$

then the moment of $\vec{F}$ about o may be calculated as:

$$M_\circ = \vec{r} \wedge \vec{F} = \frac{1}{\sqrt{14}}\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 2 & -1 \\ -3 & -1 & 2 \end{vmatrix}$$

$$= \frac{1}{\sqrt{14}}(3\underline{i} - \underline{j} + 4\underline{k})$$

2.7 Couple



Consider a pair of equal and opposite forces that are not colinear. Such a pair is called a *couple*. A couple has no resultant force. Couple is characterized by the following qualities:

1. Force F.

2. Arm of the couple which is the perpendicular distance between their lines of action.

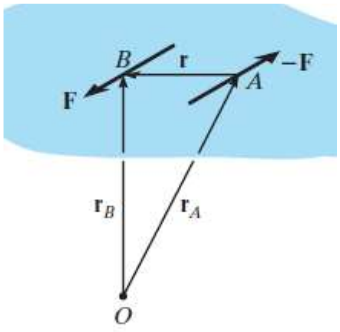
3. Plane of the couple which is the plane containing the lines action of force.

4. Moment of the couple *M* is represented by a vector directed normal to the plane of the couple and sense of *M* is determined with respect to the notation used with right hand coordinate system. So, the set of forces which their resultant vanishes but their moment cannot be ignored are called couples, in which the following conditions must be satisfied

$$\underline{F} = \sum_i \underline{F_i} = 0$$

$$\underline{M_\circ} = \sum_i \underline{r_i} \wedge \underline{F_i} \neq 0$$

The moment produced by a couple is called a ***couple moment***. We can determine its value by finding the sum of the moments of both couple forces about *any* arbitrary point. For example, in the opposite Fig., position vectors $\overrightarrow{r_A}$ and $\overrightarrow{r_B}$ are directed from point *o* to points *A* and *B* lying on the line of action of -F and F. The couple moment determined about



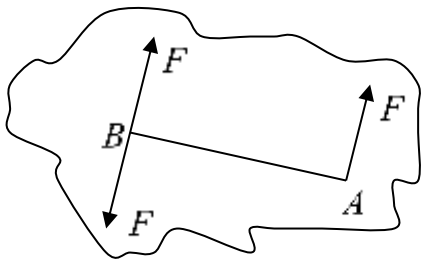
$$M_o = \overrightarrow{r_B}\, F - \overrightarrow{r_A} F = (\overrightarrow{r_B} - \overrightarrow{r_A})F$$

$$\therefore M_o = r \times F$$

This result indicates that a couple moment is a ***free vector***, i.e., it can act at *any point* since *M* depends *only* upon the position vector *r* directed *between* the forces and *not* the position vectors $\overrightarrow{r_A}$ and $\overrightarrow{r_B}$,

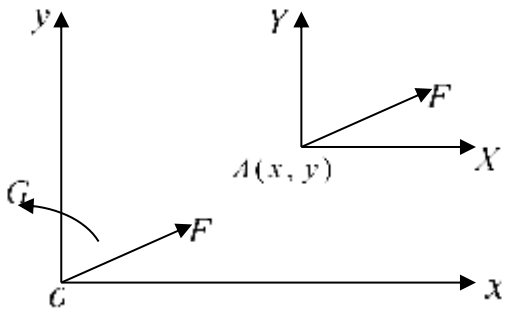
2.8 Every System of Forces is Equivalent to a Force and a Couple

Suppose that the force $\underline{F}$ affects the point A. If we wanted to move this force to another point B. To do this we add two equal forces $\underline{F}$ at point B. We must note that the adding for these two forces will not change the static conditions of the body. So $\underline{F}$ at A is equivalent for similar force at B and a couple with a moment equal to the moment of $\underline{F}$ at a point A about the point B is illustrated in the Fig.



2.9 Transport for a set of forces

If a set of plane forces effect on any arbitrary particle, so, it can be replaced at the same plane with a force that effects on the point o and a couple. Suppose that



$$\underline{F} = X\underline{i} + Y\underline{j}$$

$$\underline{oA} = \underline{r} = x\underline{i} + y\underline{j}$$

$$\therefore \ \underline{M} = \underline{r} \wedge \underline{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ x & y & 0 \\ X & Y & 0 \end{vmatrix} = (xY - yX)\underline{k}$$

$$\therefore \ |\underline{M}| = xY - yX$$

Then we observe that the set of forces are equivalent to a set of similar forces at a point o and a set of couples. The resultant for the set of forces is assumed to be R such that

(1) $$R_x = \sum X \quad , \quad R_y = \sum Y$$

Likewise, the resultant couple has a moment equal to

(2) $$M_\circ = \sum(xY - yX)$$

Also, if we try to transfer the set $(R_x, R_y, M_\circ)$ to any another point A, so, the resultant force unchanged but the moment of couple at A (M_A) will be changed as follows.

$$\underline{M_A} = \underline{M_\circ} + \underline{Ao} \wedge \underline{R}$$
$$= \underline{M_\circ} - \underline{r} \wedge \underline{R}$$

$$\therefore \ \underline{M_A} = \underline{M_\circ} - \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ x & y & 0 \\ R_x & R_y & 0 \end{vmatrix} = M_\circ \underline{k} - (xR_y - yR_x)\underline{k}$$

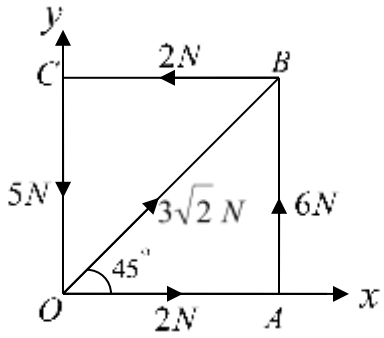
$$\therefore \ M_A = M_\circ - (xR_y - yR_x)$$

If the resultant force passes with the point A, then $M_A = 0$ and

$$M_\circ - xR_y + yR_x = 0 \tag{3}$$

Which represent the line of the resultant force.

Example 2.16 If the length of the side for a square $OABC$ is $1\ m$. The forces $2,6,2,5,3\sqrt{2}$ affected the sides OA, AB, BC, CO, OB; respectively. Find the magnitude, the direction, and the line of the resultant force.



Solution

The set of forces can be reduced to a resultant force (R_x, R_y) and a couple with a moment $M_\circ$ at o.

$$R_x = 2 + 3\sqrt{2}\,cos\,45^\circ - 2$$
$$= (3\sqrt{2})\left(\frac{1}{\sqrt{2}}\right) = 3\ N$$

$$R_y = 6 + 3\sqrt{2}\,sin\,45^\circ - 5 = 1 + (3\sqrt{2})\left(\frac{1}{\sqrt{2}}\right) = 4\ N$$

$$M_\circ = (6)(1) + 2(1) = 8\ Nm$$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{3^2 + 4^2} = 5\ N$$

$$\theta = tan^{-1}\frac{R_y}{R_x} = tan^{-1}\frac{4}{3} = 53.13^\circ$$

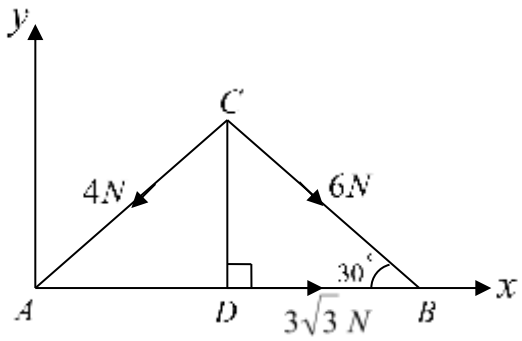
Likewise, the line of the resultant force can be written as:

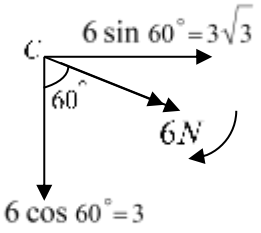
$$M_\circ - xR_y + yR_x = 0$$

$$\therefore \quad 8 - 4x + 3y = 0 \quad or \quad 4x - 3y + 8 = 0$$

Example 2.17 ABC is a triangle with $AC = BC = 8\ m,\ \hat{ACB} = 120°$. If the forces $3\sqrt{3}, 6, 4\ N$ affects the sides AB, CB, CA; respectively. Find the magnitude, direction and the line of the resultant force.

Solution





From the triangle CBD we have $BD = 4\sqrt{3}\ m$, $CD = 4\ m$, then we get

$$R_x = 3\sqrt{3} - 4\,cos\,30° + 6\,cos\,30°$$
$$= 3\sqrt{3} - 4 \cdot \frac{\sqrt{3}}{2} + 6 \cdot \frac{\sqrt{3}}{2} = 4\sqrt{3}\ N$$

$$R_y = -6\,sin\,30° - 4\,sin\,30° = -5\ N$$

$$M_\circ = -(3\sqrt{3})(CD) - 3(AD) = -(3\sqrt{3})(4) - 3(4\sqrt{3})$$
$$= -24\sqrt{3}\ Nm$$

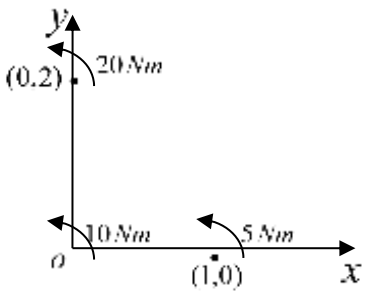
$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(4/3)^2 + (-5)^2} = \sqrt{73}\ N \approx 8.5\ N$$

$$\theta = tan^{-1}\frac{R_y}{R_x} = tan^{-1}\left(\frac{-5}{4/3}\right) = 144.18°$$

Also, the line of the resultant force can be written as:

$$M_\circ - xR_y + yR_x = 0$$
$$-24\sqrt{3} - x(-5) + y(4\sqrt{3}) = 0$$
$$i.e\ \ 5x - 4\sqrt{3}y - 24\sqrt{3} = 0$$

Example 2.18 If the algebraic sum for the moment about the points $(0,0), (1,0), (0,2)$ are $10,5,20\ Nm$. Find the magnitude, direction and the line of the resultant force.



Solution

From the following relation, we have

$$M_A = M_\circ - xR_y + yR_x$$

At the point $(0,0)$, the moment is equal to $10\ Nm$, this means that

(1) $$\therefore\ M_\circ = 10$$

At the point $(1,0)$, the moment is equal to $5\ Nm$, this means that:

(2) $$\therefore\ M_\circ - R_y = 5$$

Also, at the point $(0,2)$, the moment is equal to $20\ Nm$, this means that:

(3) $$\therefore\ M_\circ + 2R_x = 20$$

After solving these previous three equations we have

$$M_\circ = 10\ Nm \quad , \quad R_x = 5\ N \quad , \quad R_y = 5\ N$$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{5^2 + 5^2} = \sqrt{50}\ N = 5\sqrt{2}\ N$$

$$tan\,\theta = \frac{R_y}{R_x} = 1 \quad \Rightarrow \quad \therefore \theta = 45^\circ$$

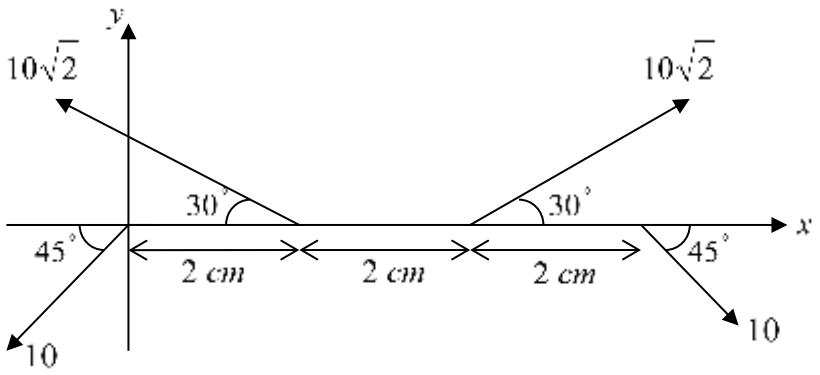
Also, the line of the resultant force can be written as:

$$M_\circ - xR_y + yR_x = 0$$

$$\therefore 10 - 5x + 5y = 0 \quad i.e \quad x - y - 2 = 0$$

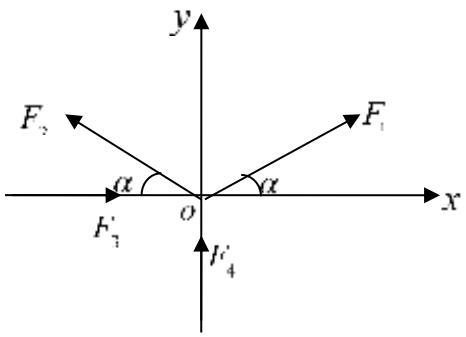
Exercise 2

1. Prove that the opposite system of forces in equilibrium



2. In the opposite Fig. Find F_3 *and* F_4 if the system is in equilibrium state and

$$F_1 = 260\ N \quad , \quad F_2 = 520\ N \quad , \quad tan\,\alpha = 5/12$$



3. Find the moment of the force $\underline{F} = 2\underline{i} - 3\underline{j}$ which effect on the point $A(1,2)$ about the origin o, and find the normal distance from o on the line of force F.

4. Three forces $\underline{F_1} = \underline{i} + \underline{j}$, $\underline{F_2} = \underline{i} - \underline{j}$, $\underline{F_3} = 2\underline{i} - 3\underline{j}$ act on a point A(1,1). Prove that the resultant moment about the point B(2,3) equal the moment of the resultant force.

5. If the algebraic sum for the moment about the points (0,4), (0,0), (2,0) are $18,12,8\ Nm$. Find the magnitude, direction and the line of the resultant force.

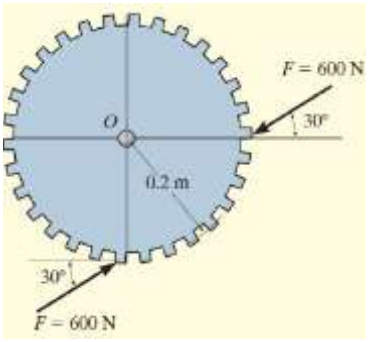
6. A rectangle $ABCD$ of side length $AD = 3\ m, AB = 4\ m$. The forces 5,4,3,6,5 affected the sides AC, DA, DC, BC, AB; respectively. Find the magnitude, the direction, and the line of the resultant force.

7. A regular hexagon $ABCDEF$ of side length $= a\ m$ and center o. The forces 1,2,3,5,4,2 affected the sides AB, BC, CD, DE, EF, FA; respectively. Find the magnitude, the direction, and the line of the resultant force.

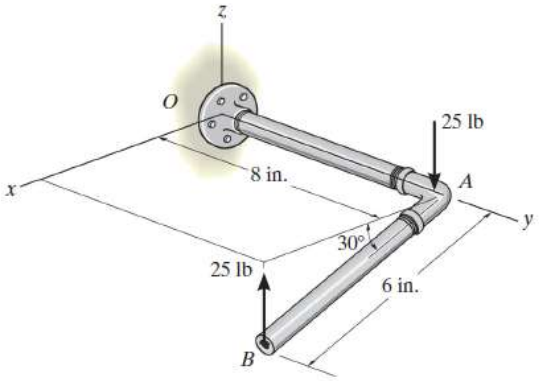
8. If the length of the side for a square $ABCD$ is $1\ ft$. The forces $2,6,2,5,3\sqrt{2}$ affected the sides AB, BC, CD, DA, BD ; respectively. Find the magnitude, the direction, and the line of the resultant force, if AB and BD are main axes.

9. Determine the moment of the force $F = 60$ passes through two points $(10,9,-3) and\ (8,8,-5)$ about the point (1,2,3).

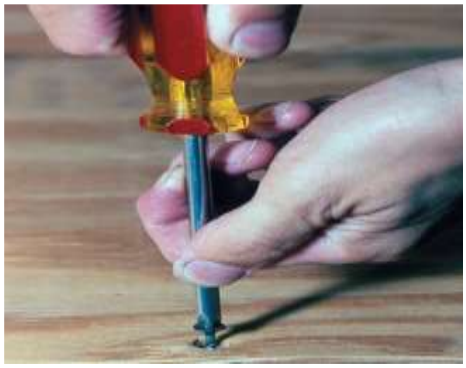
10. Determine the magnitude and direction of the couple moment acting on the gear in the opposite Fig.



11. Determine the couple moment acting on the pipe shown in the next Fig., segment *AB* is directed 30° below the *x*–*y* plane.



Chapter 3
Wrench



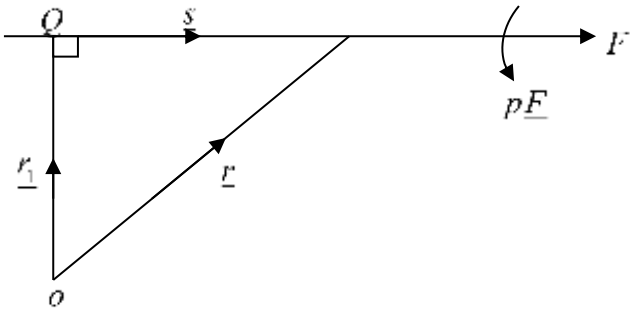
Any force system can be represented by an equivalent force and a couple at any point. But force systems can be reduced to simpler forms. That this is has more theoretical than practical importance. In this case, there is a line of points for which the system can be reduced to a force and a couple where the force, couple, and line are all parallel. The representation of the system of forces as a force and a parallel moment is called a *wrench*. This means that; if the force and the couple acting on the body are not coplanar, these can be replaced by an equivalent system which consists of a force and a couple vector lying along same line of action. This equivalent system consisting of the force F and a collinear couple CF is known as wrench. In the case of Wrench we have $\underline{F} \wedge \underline{M} = 0$.

3.1 Wrench Result

When the resultant couple vector M is parallel to the resultant force R, as shown in Fig., the resultant is called a wrench. By definition, a wrench is positive if the couple and force vectors point in the same direction and negative if they point in opposite directions.



To determine the equation of Wrench line, we assume that $\underline{r}$ is a position vector for any arbitrary point in a line of force $\underline{F}$ and $\underline{r_1}$ is a position vector for any other point Q in a line of force $\underline{F}$ in which $\underline{r_1}$ is perpendicular on $\underline{F}$ ($\underline{r_1} \cdot \underline{F} = 0$) (see the following Fig.).



So we observe that

(I) $$\underline{r} = \underline{r_1} + \underline{s} = \underline{r_1} + \lambda \underline{F}$$

But

$$\underline{M} = \underline{M_\circ} - \underline{r} \wedge \underline{F} \quad \Rightarrow \quad \therefore \underline{M_\circ} = \underline{M} + \underline{r_1} \wedge \underline{F} = p\underline{F} + \underline{r_1} \wedge \underline{F}$$

After the vector product for previous equation by $\underline{F}$ we get:

(II) $$\underline{F} \wedge \underline{M_\circ} = \underline{F} \wedge (p\underline{F} + \underline{r_1} \wedge \underline{F}) = F^2 \underline{r_1} - (\underline{F} \cdot \underline{r_1})\underline{F} = F^2 \underline{r_1}$$

$$\Rightarrow \quad \therefore \ \underline{r_1} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2}$$

which represent the wrench axis. Also, by using (I) we have:

$$\underline{r} = \underline{r_1} + \lambda \underline{F}$$

(III) $$\therefore \underline{r} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2} + \lambda \underline{F}$$

the last equation represents the central axis or the Poisson axis, which can be deduced by another way as follows:

(1) $$\underline{M} = \underline{M_\circ} - \underline{r} \wedge \underline{F}$$

$$\therefore \ \underline{F} \wedge (\underline{M_\circ} - \underline{r} \wedge \underline{F}) = 0$$

$$\underline{F} \wedge \underline{M_\circ} - \underline{F} \wedge (\underline{r} \wedge \underline{F}) = 0$$

$$\Rightarrow \underline{F} \wedge \underline{M_\circ} - F^2 \underline{r} + (\underline{F} \cdot \underline{r}) = 0$$

(2) $$\Rightarrow \underline{r} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2} + \lambda \underline{F}$$

Where λ a scalar quantity is has the following value.

(3) $$\lambda = \frac{\underline{F} \cdot \underline{r}}{F^2}$$

By using (2) in (1) we get:

$$\underline{M} = \underline{M_\circ} - \left(\frac{\underline{F} \wedge \underline{M_\circ}}{F^2} + \lambda \underline{F} \right) \wedge \underline{F}$$

(4) $$\underline{M} = \frac{\underline{F}\cdot\underline{M}_\circ}{F^2}\ \underline{F}$$

3.2 Wrench Degree

From the last equation (4) we note that

(5) $$\frac{\underline{M}}{\underline{F}} = \frac{\underline{F}\cdot\underline{M}_\circ}{F^2} = p$$

Where p is called the *Wrench degree*. Likewise, to get the Cartesian form for the equation of the Wrench axis we have:

$$x\underline{i}+y\,\underline{j}+z\underline{k}=\frac{1}{F^2}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ F_x & F_y & F_z \\ M_x & M_y & M_z \end{vmatrix}+\lambda(F_x\underline{i}+F_y\,\underline{j}+F_z\underline{k})$$

By eliminating the parameter λ from the last equation we get:

$$\frac{x-\frac{(M_zF_y-M_yF_z)}{F^2}}{F_x}=\frac{y-\frac{(M_xF_z-M_zF_x)}{F^2}}{F_y}=\frac{z-\frac{(M_yF_x-M_xF_y)}{F^2}}{F_z}$$

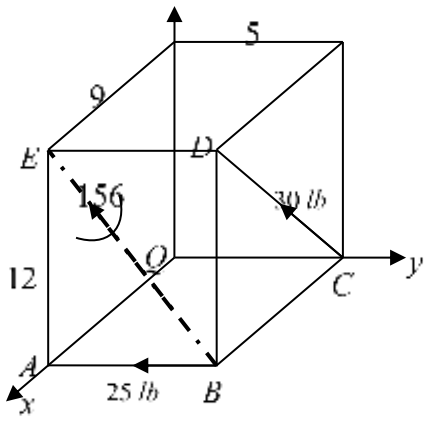
By comparing this equation with the equation of the straight line in the space we must observe that:

$$\frac{x-x_\circ}{l}=\frac{y-y_\circ}{m}=\frac{z-z_\circ}{n}=\mu$$

The effected point $(x_\circ, y_\circ, z_\circ)$ for the Wrench after $\mu = 0$ is

$$x_\circ = \frac{(M_zF_y-M_yF_z)}{F^2}$$
$$y_\circ = \frac{(M_xF_z-M_zF_x)}{F^2}$$
$$z_\circ = \frac{(M_yF_x-M_xF_y)}{F^2}$$

Example 3.1 In the following figure, find the equivalent wrench for the forces and couples.



Solution

Firstly we note that

$$\underline{F_1} = -25\ \underline{j} \quad lb \text{ acts on the point } \underline{r_1} = 9\ \underline{i}$$

But the magnitude of $\underline{F_2}$ which is equal to $30\ lb$ effected on the direction $\overrightarrow{CD}$, so we must deduce a unit vector in this direction as follows:

$$\overset{\wedge}{CD} = \frac{9\underline{i} + 12\underline{k}}{\sqrt{81 + +144}} = \frac{1}{5}(3\underline{i} + 4\underline{k})$$

$$\therefore \underline{F_2} = 6(3\underline{i} + 4\underline{k})$$

Which effect on the following point,

$$\underline{r_2} = 5\ \underline{j}$$

Likewise, the couple with moment 156 effect on the direction $\overrightarrow{BE}$. So we must deduce a unit vector in the direction of $\overrightarrow{BE}$ as follows:

$$\overset{\wedge}{BE} = \frac{-5\underline{j} + 12\underline{k}}{\sqrt{25 + +144}} = \frac{1}{13}(-5\underline{i} + 12\underline{k})$$

So,

$$\underline{C_1} = 12(-5\underline{j} + 12\underline{k}) \quad lb.ft$$

Also,

$$\underline{F} = \underline{F_1} + \underline{F_2} = 18\underline{i} - 25\underline{j} + 24\underline{k} \quad lb$$

In the same time after getting the moment about Q for the set $(\underline{F_1}, \underline{F_2}, \underline{C_1})$ we notice that:

$$\underline{M_\circ} = \underline{C_1} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -25 & 0 & 0 \\ 0 & 9 & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 5 & 0 \\ 10 & 0 & 24 \end{vmatrix}$$

$$\underline{M_\circ} = 120\underline{i} - 60\underline{j} - 171\underline{k}$$

Also, the degree of the wrench can be calculated by the following equation:

$$p = \frac{1}{F^2}(\underline{F} \cdot \underline{M_\circ}) = \frac{18 \times 20 + 60 \times 25 - 171 \times 45}{(18)^2 + (25)^2 + (24)^2}$$

$$= -\frac{444}{1525} = -0.3\ ft$$

With the direction

$$\underline{r} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2} = \frac{5455\underline{i} + 5778\underline{j} + 720\underline{k}}{1525}$$
$$= 3.6\underline{i} + 3.8\underline{j} + 0.5\underline{k}$$

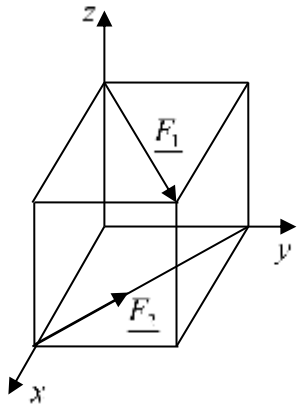
Finally, we have:

$$\underline{F} = 18\underline{i} - 25\underline{j} + 24\underline{k} \quad lb$$

$$\underline{r} = 3.6\underline{i} + 3.8\underline{j} + 0.5\underline{k}$$

$$\underline{M} = -0.3\underline{F}$$

Example 3.2 In the following figure, two forces F effects on the cube with side length a. Find the equivalent wrench for the following system.



Solution

$$\underline{F_1} = \frac{\sqrt{2}}{2}F(\underline{i} + \underline{j}) \quad , \quad \underline{r_1} = a\underline{k}$$

$$\underline{F_2} = \frac{\sqrt{2}}{2}F(-\underline{i} + \underline{j}) \quad , \quad \underline{r_2} = a\underline{i}$$

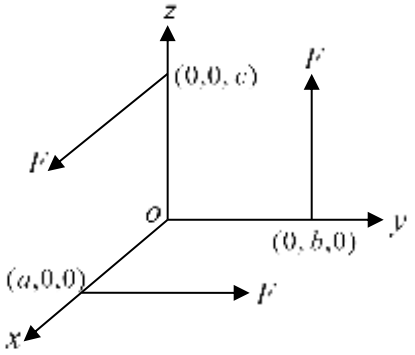
(i) $$\underline{R} = \underline{F_1} + \underline{F_2} = \sqrt{2}F\underline{j}$$

(ii) $$\underline{M_\circ} = \underline{r_1} \wedge \underline{F_1} + \underline{r_2} \wedge \underline{F_2} = \frac{aF}{\sqrt{2}}(-\underline{i} + \underline{j} + \underline{k})$$

(iii) $$p = \frac{1}{F^2}(\underline{F} \cdot \underline{M_\circ}) = a$$

(iv) $$\underline{r} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2} = a(\underline{i} + \underline{k})$$

Example 3.3 Three equal forces each of them is equal to F. The first one effect on the point $(a, 0, 0)$ in the parallel direction to oy axis, the second effect on $(0, b, 0)$ in the parallel direction to oz axis and the final force effect on $(0, 0, c)$ in the parallel direction to ox axis. Find the equivalent wrench for the following forces.



Solution

The resultant force $\underline{F} = (F, F, F)$

Also, the moment of the couple is equal to:

$$\underline{M_\circ} = \underline{r_1} \wedge \underline{F_1} + \underline{r_2} \wedge \underline{F_2} + \underline{r_3} \wedge \underline{F_3}$$

$$= bF\underline{i} + cF\underline{j} + aF\underline{k} = F(b\underline{i} + c\underline{j} + a\underline{k})$$

Likewise, the degree of the wrench is equal to :

$$p = \frac{1}{F^2}(\underline{F} \cdot \underline{M_\circ}) = \frac{1}{3}(a + b + c)$$

Whereas the equation of the wrench axis is:

$$\underline{r} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2} + \lambda \underline{F}$$

and in the Cartesian coordinates we get:

$$x\underline{i} + y\underline{j} + z\underline{k} = \frac{1}{3F^2}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ F & F & F \\ bF & cF & aF \end{vmatrix} + \lambda(F\underline{i} + F\underline{j} + F\underline{k})$$

Then,

$$x = \frac{1}{3}(a - c) + \lambda F \quad \Rightarrow \quad 3x - a + c = \lambda F$$

$$y = \frac{1}{3}(b - a) + \lambda F \quad \Rightarrow \quad 3y - b + a = \lambda F$$

$$z = \frac{1}{3}(c - a) + \lambda F \quad \Rightarrow \quad 3z - c + a = \lambda F$$

Finally, we have:

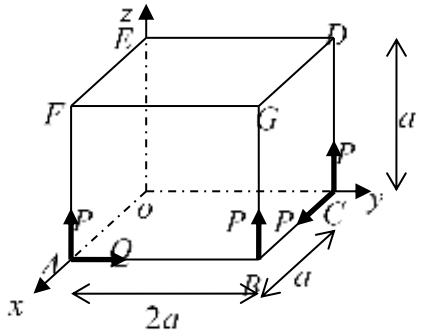
$$3x - a + c = 3y - b + a = 3z - c + a$$

Example 3.4 Four forces of magnitude P effect on the opposite cuboid. Find the value of the force Q which make the system equivalent wrench and find its equation of axis.

Solution

$$\underline{F} = P\underline{i} + Q\underline{j} + 3P\underline{k}$$

$$\underline{M_o} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a & 0 & 0 \\ 0 & 0 & P \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a & 0 & 0 \\ 0 & Q & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a & 2a & 0 \\ 0 & 0 & P \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 2a & 0 \\ 0 & 0 & P \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 2a & 0 \\ P & 0 & 0 \end{vmatrix} = 4aP\underline{i} - 2aP\underline{j} + a(Q-2P)\underline{k}$$



This equivalent to a force, if

$$\underline{F} \cdot \underline{M_o} = 0 \quad \Rightarrow \quad \therefore \; 4aP^2 - 2aPQ + 3aP(Q-2P) = 0 \quad \Rightarrow \quad \therefore aPQ = 2aP^2 \quad \Rightarrow \quad \therefore Q = 2P$$

where, the line equation

$$M_{ox} = yR_z - zR_y \quad , \quad M_{oy} = zR_x - xR_z \quad , \quad M_{oz} = xR_y - yR_x$$
$$4aP = 3Py - 2Pz \quad , \quad -2aP = Pz - 3Px \quad , \quad 0 = 2Px - Py$$

$$\therefore 3y - 2z = 4a \quad , \quad 3x - z = 2a \quad , \quad 2x - y = 0$$

Example 3.5 In the previous example if $Q = P,$ prove that the system equivalent to wrench and find its equation of axis.

Solution

if $Q = P$

$$\underline{R} = P\left(\underline{i} + \underline{j} + 3\underline{k}\right) \quad \Rightarrow \quad R_x = P \; , \; R_y = P \; , \; R_z = 3P, \; \left|\underline{R}\right| = \sqrt{11}P$$
$$\underline{M_\circ} = aP(4\underline{i} - 2\underline{j} - 3\underline{k}) \Rightarrow \quad M_{ox} = 4aP \; , \; M_{oy} = -2aP \; , \; M_{oz} = -3aP$$

$$\underline{R} \cdot \underline{M_\circ} = aP^2(4 - 2 - 9) = -7a\,P^2$$

where $\underline{R} \cdot \underline{M_o} \neq 0$ it means that the system equivalent to Wrench.

$$M_\circ \cos\theta = \frac{\underline{R} \cdot \underline{M_\circ}}{R} = \frac{aP^2(4-2-9)}{\sqrt{11}P} = -\frac{7}{\sqrt{11}}aP$$

then, the Wrench degree

$$p = \frac{M_\circ \cos\theta}{R} = -\frac{7}{\sqrt{11}}a$$

wrench axis in the Cartesian coordinates.

$$\underline{r_{o'}} = x'\underline{i} + y'\underline{j} + z'\underline{k} = \frac{\underline{R} \wedge \underline{M_\circ}}{R^2} = \frac{1}{R^2}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ R_x & R_y & R_z \\ M_{ox} & M_{oy} & M_{oz} \end{vmatrix} = \frac{1}{11P^2}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ P & P & 3P \\ aP & -2aP & -3aP \end{vmatrix}$$

$$= \frac{a}{11}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 3 \\ 1 & -2 & -3 \end{vmatrix} \Rightarrow \quad \therefore \; x' = \frac{3}{11}a, \quad y' = 0, \quad z' = -\frac{3}{11}a \Rightarrow o' \equiv \left(\frac{3a}{11}, 0, -\frac{3a}{11}\right)$$

$$\therefore \; \frac{x - (3/11)a}{P} = \frac{y - 0}{P} = \frac{z + (3/11)a}{3P}$$

$$x - \frac{3}{11}a = y = \frac{1}{3}z + \frac{1}{11}a$$

Example 3.6 A set of forces equivalent to a force at the origin $\underline{F} = f(6\underline{i} - 6\underline{j} + 2\underline{k})$ and a moment $\underline{M_\circ} = f(-2\underline{i} - 5\underline{j} + 10\underline{k})$ Find the equivalent Wrench and its equation of axis.

Solution

The Wrench degree

$$P = \frac{\underline{F} \cdot \underline{M_\circ}}{F^2} = \frac{f^2(-12 + 30 + 20)}{f^2(36 + 36 + 4)} = \frac{38}{76} = \frac{1}{2}$$

the axis equation can be found from the relation

$$\underline{r} = \frac{\underline{F} \wedge \underline{M_\circ}}{F^2} + \lambda\underline{F} = \frac{1}{76f^2}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 6f & -6f & 2f \\ -2f & -5f & 10f \end{vmatrix} + \lambda f(6\underline{i} - 6\underline{j} + 2\underline{k})$$

by equating coefficient of $\underline{i}, \underline{j}, \underline{k}$, for both sides

$$x = \frac{1}{76f^2}(-50f^2) + 6\lambda f = -\frac{25}{38} + 6\lambda f \Rightarrow \lambda f = \frac{x}{6} + \frac{25}{228}$$
$$y = \frac{1}{76f^2}(-64f^2) - 6\lambda f = -\frac{16}{19} - 6\lambda f \Rightarrow \lambda f = -\frac{y}{6} - \frac{8}{57}$$
$$z = \frac{1}{76f^2}(-42f^2) + 2\lambda f = -\frac{21}{38} + 2\lambda f \Rightarrow \lambda f = \frac{z}{2} + \frac{21}{76}$$

then

$$\frac{x}{6} + \frac{25}{228} = -\frac{y}{6} - \frac{8}{57} \Rightarrow \frac{x}{6} + \frac{y}{6} = -\frac{1}{4}$$

$$\therefore \quad x + y = -\frac{3}{2}$$

$$\frac{x}{6}+\frac{25}{228}=\frac{z}{2}+\frac{21}{76}\Rightarrow\frac{x}{6}-\frac{z}{2}=\frac{1}{6}$$

$$\therefore\quad x-3z=1$$

$$-\frac{y}{6}-\frac{8}{57}=\frac{z}{2}+\frac{21}{76}\Rightarrow\frac{y}{6}+\frac{z}{2}=-\frac{5}{12}$$

$$\therefore\quad y+3z=-\frac{5}{2}$$

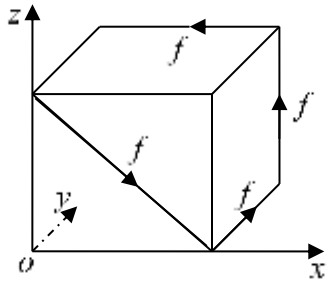
from the previous equation

$$x+y=-\frac{3}{2}\quad,\quad x-3z=1\quad,\quad y+3z=-\frac{5}{2}$$

$$\therefore\ \frac{x-1}{3}=\frac{y+5/2}{-3}=z$$

which is the Cartesian equation of Wrench axis.

Example 3.7 In the following figure, four forces f effect on the cube with side length l. Find the resultant force and moment about the origin *o* and determine which it equivalent to a wrench or a unique force.



Solution

We have four forces:

1. Two forces $\underline{f_1}=f\ \frac{\underline{i}-\underline{k}}{\sqrt{2}}\ and\ \underline{f_2}=f\underline{j}$ effect on point $(l,0,0)$

2. Two forces $\underline{f_3}=f\underline{k}\ \ and\ \ \underline{f_4}=-f\underline{i}$ effect on point (l,l,l)

then, the resultant force is $\underline{F}=f\left[\underline{i}\ \left(\frac{1}{\sqrt{2}}-1\right)+\underline{j}+\underline{k}\ \left(-\frac{1}{\sqrt{2}}+1\right)\right]$, the moment about the origin

$$\underline{M_\circ}=\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\ l&0&0\\ \frac{f}{\sqrt{2}}&0&-\frac{f}{\sqrt{2}}\end{vmatrix}+\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\ l&0&0\\ 0&f&0\end{vmatrix}+\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\ l&l&l\\ 0&0&f\end{vmatrix}+\begin{vmatrix}\underline{i}&\underline{j}&\underline{k}\\ l&l&l\\ -f&0&0\end{vmatrix}$$

$$\therefore\ \underline{M_\circ}=fl\underline{i}+\left(\frac{fl}{\sqrt{2}}-2fl\right)\underline{j}+2fl\underline{k}=fl\left[\underline{i}+\left(\frac{1}{\sqrt{2}}-2\right)\underline{j}+2\underline{k}\right]$$

the Wrench degree

$$P = \frac{\underline{F} \cdot \underline{M_\circ}}{F^2}$$

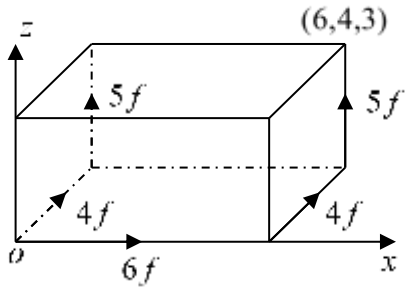
$$\underline{F} \cdot \underline{M_\circ} = f^2 l\left[\frac{1}{\sqrt{2}} - 1 + \frac{1}{\sqrt{2}} - 2 + 2\left(-\frac{1}{\sqrt{2}} + 1\right)\right] = -f^2 l$$

$$F^2 = \left[\left(\frac{1}{\sqrt{2}} - 1\right)^2 + 1^2 + \left(-\frac{1}{\sqrt{2}} - 1\right)^2\right] f^2 = (4 - 2\sqrt{2}) f^2$$

where $\underline{F} \cdot \underline{M_\circ} \neq 0$, then the system equivalent to Wrench of degree

$$P = \frac{-f^2 l}{(4 - 2\sqrt{2}) f^2} = -\frac{l}{2\sqrt{2}(\sqrt{2} - 1)}$$

Example 3.8 In the following cuboid, five forces effect on it. determine which the system is equivalent to a wrench or a unique force and find its equation of axis.



Solution

There are five forces:

1. Two forces $\underline{f_1} = 6f\underline{i}$, $\underline{f_2} = 4f\underline{j}$ effect on the point (0,0,0)

2. The force $\underline{f_3} = 4f\ \underline{j}$ effects on point (6,0,0)

3. The force $\underline{f_4} = 5f\ \underline{k}$ effects on point (6,4,0)

4. The force $\underline{f_5} = 5f\ \underline{k}$ effects on point (0,4,0)

then, the resultant force $\underline{F} = f\left[6\underline{i} + 4\underline{j} + 4\underline{j} + 5\underline{k} + 5\underline{k}\right] = f(6\underline{i} + 8\underline{j} + 10\underline{k})$

The moment about *o*

$$\underline{M_\circ} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 0 & 0 \\ 6f & 0 & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 0 & 0 \\ 0 & 4f & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 6 & 0 & 0 \\ 0 & 4f & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 6 & 4 & 0 \\ 0 & 0 & 5f \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 4 & 0 \\ 0 & 0 & 5f \end{vmatrix}$$

$$\therefore\ \underline{M_\circ} = f(40\underline{i} - 30\underline{j} + 24\underline{k})$$

$$\underline{F} \cdot \underline{M_\circ} = f^2[240 - 240 + 240] = 240f^2$$

$$F^2 = f^2(36 + 64 + 100) = 200f^2$$

The Wrench degree

$$\therefore P = \frac{\underline{F} \cdot \underline{\underline{M_\circ}}}{F^2} = \frac{240}{200} = \frac{6}{5}$$

We obtain the equation of axis

$$\underline{r} = \frac{\underline{F} \wedge \underline{\underline{M_\circ}}}{F^2} + \lambda \underline{F} = \frac{1}{200f^2}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 6f & 8f & 10f \\ 40f & -30f & 24f \end{vmatrix} + \lambda f(6\underline{i} + 8\underline{j} + 10\underline{k})$$

$$x = \frac{492}{200} + 6\lambda f = \frac{123}{50} + 8\lambda f \Rightarrow \frac{1}{6}\left(x - \frac{123}{50}\right) = \lambda f$$
$$y = \frac{256}{200} + 8\lambda f = \frac{32}{25} + 8\lambda f \Rightarrow \frac{1}{8}\left(y - \frac{32}{25}\right) = \lambda f$$
$$z = -\frac{500}{200} + 10\lambda f = -\frac{5}{2} + 10\lambda f \Rightarrow \frac{1}{10}(z + \frac{5}{2}) = \lambda f$$

Then, the equation of axis

$$\frac{1}{6}\left(x - \frac{123}{50}\right) = \frac{1}{8}\left(y - \frac{32}{25}\right) = \frac{1}{10}\left(z + \frac{5}{2}\right)$$

Example 3.9 Consider three forces f_1, f_2, f_3 effect on the straight lines

$$L_1(y = b\ ,\ z = -c)\ \ ,\ \ L_2(x = -a\ ,\ z = c)\ \ ,\ \ L_3(x = a\ ,\ y = -b)$$

with positive directions of coordinates *oxyz*. Prove that the system is equivalent to a unique force if

$$\frac{a}{f_1} + \frac{b}{f_2} + \frac{c}{f_3} = 0$$

and then $\quad \frac{y}{f_2} - \frac{z}{f_3} = \frac{a}{f_1} \quad , \quad \frac{z}{f_3} - \frac{x}{f_1} = \frac{b}{f_2} \quad , \quad \frac{x}{f_1} - \frac{y}{f_2} = \frac{c}{f_3}$

Solution

In this example, there are three forces

$$\underline{f_1} = f_1\underline{i} \quad , \quad (0, b, -c) \quad , \quad \underline{r_1} = b\underline{j} - c\underline{k}$$
$$\underline{f_2} = f_2\underline{j} \quad , \quad (-a, 0, c) \quad , \quad \underline{r_2} = -a\underline{i} + c\underline{k}$$
$$\underline{f_3} = f_3\underline{k} \quad , \quad (a, -b, 0) \quad , \quad \underline{r_3} = a\underline{i} - b\underline{j}$$

then, the resultant force $\quad \underline{F} = f_1\underline{i} + f_2\underline{j} + f_3\underline{k}$

and the moment about the origin

$$\underline{M_\circ} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & b & -c \\ f_1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -a & 0 & c \\ 0 & f_2 & 0 \end{vmatrix} + \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a & -b & 0 \\ 0 & 0 & f_3 \end{vmatrix}$$

$$\underline{M_\circ} = -(cf_2 + bf_3)\underline{i} - (af_3 + cf_1)\underline{j} - (af_2 + bf_1)\underline{k}$$

$$\because \underline{F} \cdot \underline{M_\circ} = -f_1(cf_2 + bf_3) - f_2(af_3 + cf_1) - f_3(af_2 + bf_1)$$
$$= -2[af_2f_3 + bf_1f_3 + cf_1f_2]$$

the system is equivalent to a unique force if $\underline{F} \cdot \underline{M_\circ} = 0$

$$\therefore [af_2f_3 + bf_1f_3 + cf_1f_2] = 0$$
$$f_1f_2f_3\left(\frac{a}{f_1} + \frac{b}{f_2} + \frac{c}{f_3}\right) = 0 \Rightarrow \therefore \frac{a}{f_1} + \frac{b}{f_2} + \frac{c}{f_3} = 0 \qquad \text{\#QED first}$$

The equation of axis can be obtained from equation

$$\underline{M_\circ} - \underline{r} \wedge \underline{F} = 0 \Rightarrow \therefore \underline{M_\circ} = \underline{r} \wedge \underline{F}$$
$$\underline{M_\circ} = -(cf_2 + bf_3)\underline{i} - (af_3 + cf_1)\,\underline{j} - (af_2 + bf_1)\,\underline{k}$$
$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ x & y & z \\ f_1 & f_2 & f_3 \end{vmatrix} = (yf_3 - zf_2)\underline{i} + (zf_1 - xf_3)\,\underline{j} + (xf_2 - yf_1)\underline{k}$$

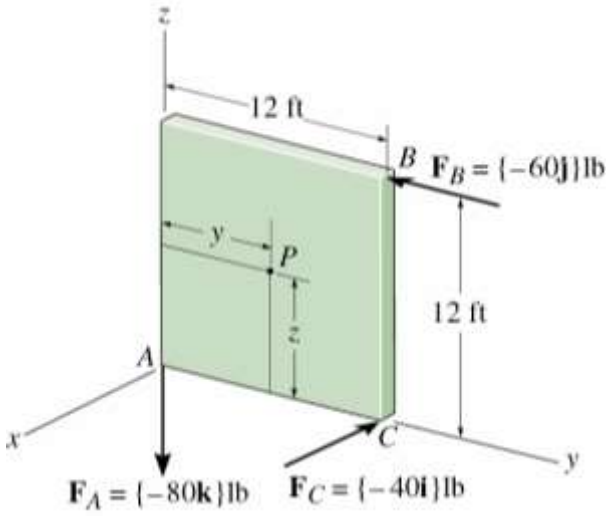
$$\therefore \quad yf_3 - zf_2 = -(cf_2 + bf_3) \qquad (1)$$
$$zf_1 - xf_3 = -(af_3 + cf_1) \qquad (2)$$
$$xf_2 - yf_1 = -(af_2 + bf_1) \qquad (3)$$

by divided equations (1), (2) and (3) by f_2f_3 , f_1f_3 , f_1f_2 respectively

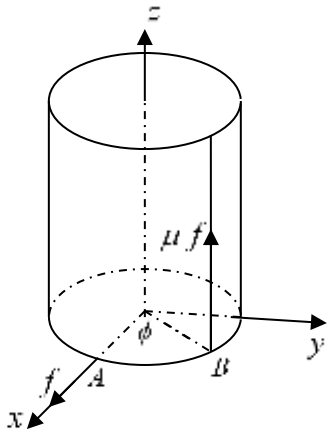
$$\frac{y}{f_2} - \frac{z}{f_3} = -\left(\frac{c}{f_3} + \frac{b}{f_2}\right) = \frac{a}{f_1}$$
$$\frac{z}{f_3} - \frac{x}{f_1} = -\left(\frac{a}{f_1} + \frac{c}{f_3}\right) = \frac{b}{f_2}$$
$$\frac{x}{f_1} - \frac{y}{f_2} = -\left(\frac{a}{f_1} + \frac{b}{f_2}\right) = \frac{c}{f_3} \qquad \text{\#QED second}$$

Exercise

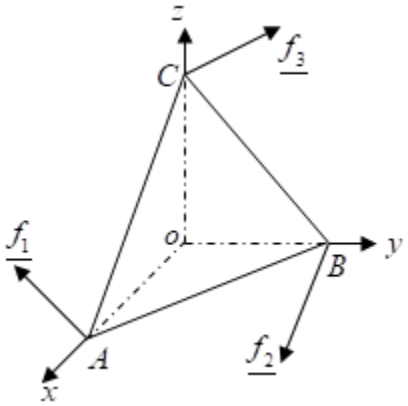
1- Find the resultant force and moment about the origin *o* and determine which it equivalent to a wrench or a unique force.



2- In the following cylinder, a force f effects on x-axis and another μf effects on one of its mapping, determine which the system is equivalent to a wrench or a unique force and find its equation of axis.



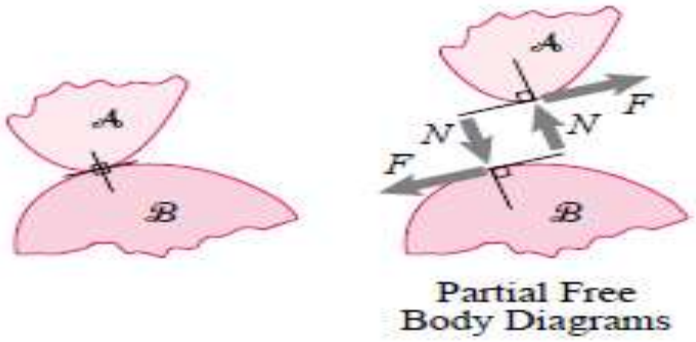
3- Let three points A, B and C all of them have a distance from the origin a. Three forces of magnitude $f_1 = f$, $f_2 = 2f, f_3 = 3f$ effect on the directions $\underline{BC}, \underline{CA}, \underline{AB}$ respectively. Find the resultant force and moment about the origin *o* and determine which it equivalent to a wrench or a unique force.



Chapter 4
Friction

Friction

When two objects are in contact and one is sliding with respect to the other, we call the force which resists this sliding *friction*. Frictional contact is usually assumed to be either 'lubricated' or 'dry.' When bodies are in lubricated contact they are not in real contact at all, a thin layer of liquid or gas separates them. Most of the metal to metal contact in a car engine is supposed to be lubricated. The contact of the car tires with the road is 'dry' unless the car is 'hydroplaning' on worn-smooth tires on a very wet road. The friction forces in lubricated contact are very small compared forces of unlubricated contact. For many purposes lubricated friction forces are neglected. There is no quick way to estimate these small lubricated slip forces. The accurate estimation of lubricated friction forces requires use of lubrication theory, a part of fluid mechanics. We will drop the discussion of lubricated friction forces because they are often negligible and because estimating them is too hard. Dry friction forces are not small and thus cannot be sensibly neglected in dynamics problems involving sliding contact. The simplest model for friction forces is called *Coulomb's law of friction* or just *Coulomb friction*. But, use of even this law is full of subtleties.



Partial Free Body Diagrams

Rough and smooth

People sometimes use the words *rough* and *smooth* to denote dry friction and lubricated friction. This language seems to make sense because, when wet, very smooth things get slippery. However, despite the dominance of this intuition, in fact dry solids do not generally have less friction when polished than when rough.

Laws of Friction:

(i) Friction acts in the direction opposite to what would be the direction of relative motion if the friction did not exist; or, in the case of relative motion, the friction opposes the relative motion.

(ii) The magnitude of the friction is always just sufficient to preserve equilibrium or to prevent relative motion of the points of contact, provided that sufficient friction can be obtained. There is a limit, however, to the amount of friction

that can be called into play; when this limit is reached the friction is called limiting friction, and limiting friction is controlled by the following further laws:

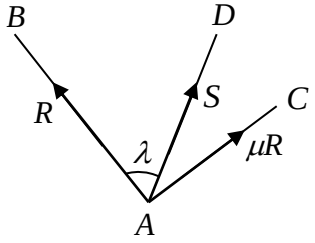
(iii) Limiting friction between two bodies bears a constant ratio to the normal reaction between them; i.e. if R denotes the normal reaction, then μR denotes the limiting friction, where μ is a constant depending on the nature of the materials in contact and called the coefficient of friction.

(iv) Limiting friction is independent of the areas in contact.

(v) When motion takes place the foregoing laws of limiting friction are still true, but the coefficient of friction μ is slightly less for bodies in motion than for the same bodies at rest; and the friction is independent of the velocity.

Angle of Friction

When two bodies in contact at a point are in limiting equilibrium so that the greatest possible amount of friction is called into play, the angle which the resultant reaction makes with the common normal at the point of contact is called the angle of friction.



If R denotes the normal reaction, then μR denotes the limiting friction and, if λ is the angle of friction,

$$S\cos\lambda = R \quad , \quad S\sin\lambda = \mu R$$

So,

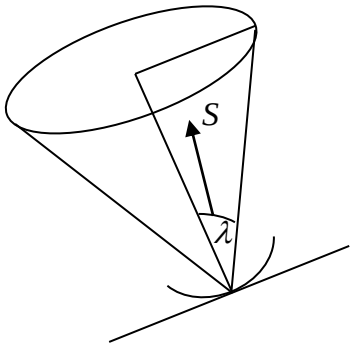
$$S = R\sqrt{1+\mu^2}$$

Then,

$$\mu = \tan\lambda$$

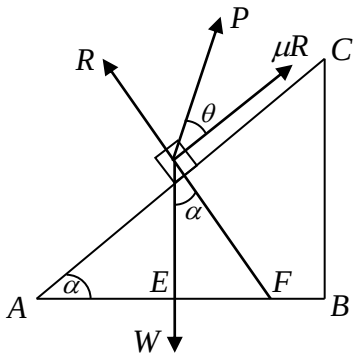
Cone of Friction

It follows that, if, with the common normal to the surfaces as axis, we construct a cone of semivertical angle λ, the angle of friction, then if the resultant of the other forces acting on either body falls within or along the surface of this cone it can be balanced by the reaction of the other body; but, if it falls without the cone, equilibrium is not possible because there is not enough friction available to balance the tangential component of the other forces acting on either body. This cone is called **the cone of friction.**



Equilibrium at an inclined rough plane

If a body of weight W is placed on an inclined plane whose inclination α is greater than the angle of friction; to find the least force which will prevent the body from sliding down the plane. Let R be the resultant reaction of the plane on the body. Since the body tends to slide down the plane the friction on the body acts up the plane. Also, let P be a force which will balance R and W, and let P make an angle θ with the plane.



Then, the eequilibrium equations can be written in the following form:

$$P\cos\theta + \mu R = W\sin\theta \qquad (1)$$

$$P\sin\theta + R = W\cos\theta \qquad (2)$$

So,

$$P = \frac{W(\sin\alpha - \mu\cos\alpha)}{\cos\theta - \mu\sin\theta}$$

But $\mu = \tan\lambda$, then

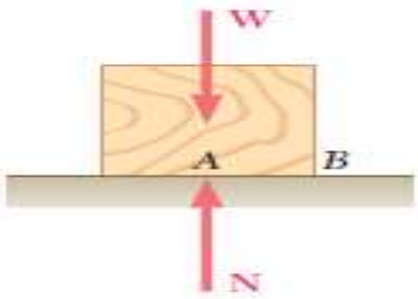
$$P = \frac{W \sin(\alpha - \lambda)}{\cos(\theta + \lambda)} \quad (3)$$

Likewise, if the body is impending sliding, then the sign for the λ changed from the positive to negative, so:

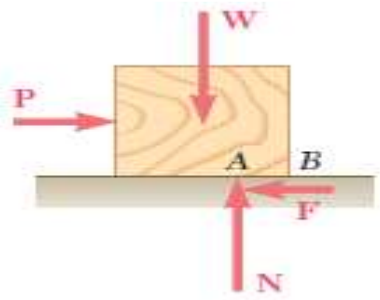
$$P_1 = \frac{W \sin(\alpha + \lambda)}{\cos(\theta - \lambda)} \quad (4)$$

Coefficients of Friction:

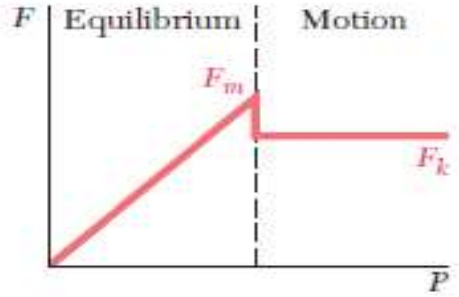
The laws of friction are exemplified by the following experiment. A block of weight **W** is placed on a horizontal plane surface (see the following Figure). The forces acting on the block are its weight **W** and the reaction of the surface. Since the weight has no horizontal component, the reaction of the surface also has no horizontal component; the reaction is therefore *normal* to the surface and is represented by **N** in the following Figure.



Suppose, now, that a horizontal force **P** is applied to the block (see the following Figure). If **P** is small, the block will not move; some other horizontal force must therefore exist, which balances **P**.



This other force is the *static-friction force* **F**, which is actually the resultant of a great number of forces acting over the entire surface of contact between the block and the plane. The nature of these forces is not known exactly, but it is generally assumed that these forces are due to the irregularities of the surfaces in contact and, to a certain extent, to molecular attraction. If the force P is increased, the friction force F also increases, continuing to oppose P, until its magnitude reaches a certain maximum value Fm (see the following Figure). If P is further increased, the friction force cannot balance it any more and the block starts sliding. As soon as the block has been set in motion, the magnitude of F drops from F_m to a lower value F_k. This is because there is less interpenetration between the irregularities of the surfaces in contact when these surfaces move with respect to each other. From then on, the block keeps sliding with increasing velocity while the friction force, denoted by F_k and called the kinetic-friction force, remains approximately constant.



Experimental evidence shows that the maximum value F_m of the static-friction force is proportional to the normal component N of the reaction of the surface. We have:

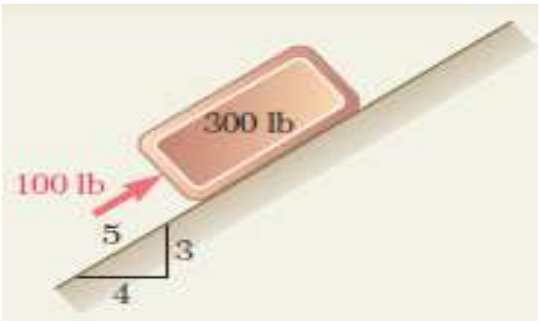
$$F_m = \mu_s N,$$

where μ_s is a constant called the coefficient of static friction. Similarly, the magnitude F_k of the kinetic-friction force may be put in the form:

$$F_k = \mu_k N,$$

where μ_k is a constant called the coefficient of kinetic friction. The coefficients of friction μ_s and μ_k do not depend upon the area of the surfaces in contact. Both coefficients, however, depend strongly on the nature of the surfaces in contact.

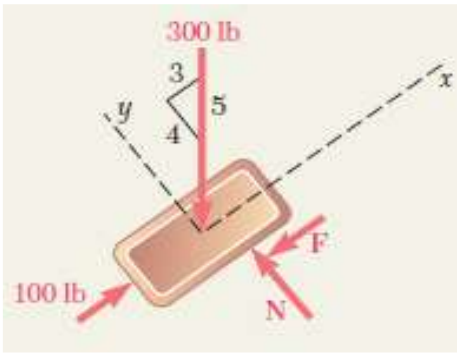
Example1: A 100-lb force acts as shown on a 300-lb block placed on an inclined plane. The coefficients of friction between the block and the plane are $\mu_s = 0.25$ and $\mu_k = 0.20$.



Determine whether the block is in equilibrium, and find the value of the friction force.

Solution:

We first determine the value of the friction force required to maintain equilibrium. Assuming that F is directed down and to the left, we draw the free-body diagram of the block and write



$$\sum F_x = 0: \quad 100 - \frac{3}{5}(300) - F = 0$$

$$\Rightarrow \qquad F = -80\,lb \Rightarrow \qquad F = 80\,lb$$

$$\sum F_y = 0: \quad N - \frac{4}{5}(300) = 0$$

$$\Rightarrow \qquad N = 240\ lb$$

The force F required to maintain equilibrium is an 80 lb force directed up and to the right; the tendency of the block is thus to move down the plane.

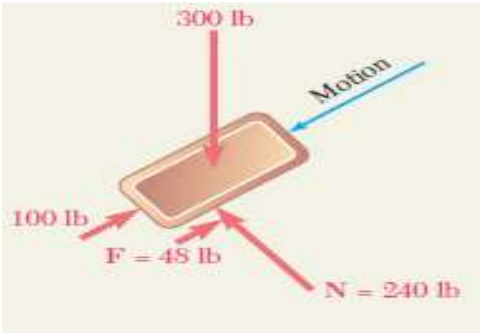
Maximum Friction Force. The magnitude of the maximum friction force which may be developed is

$$F_m = \mu_s N$$
$$\Rightarrow F_m = 0.25(240) = 60 lb$$

Since the value of the force required to maintain equilibrium (80 lb) is larger than the maximum value which may be obtained (60 lb), equilibrium will not be maintained and the block will slide down the plane.

Actual Value of Friction Force. The magnitude of the actual friction force is obtained as follows:

$$F_k = \mu_k N$$
$$\Rightarrow F_k = 0.20(240) = 48 lb$$



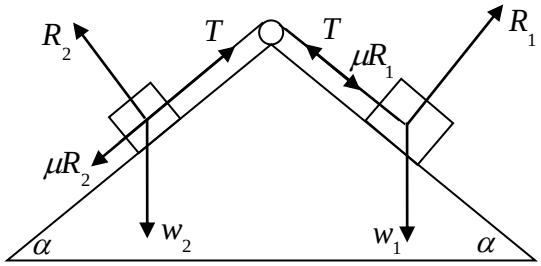
The sense of this force is opposite to the sense of motion; the force is thus directed up and to the right.

Example2: Two blocks w_1, w_2 are placed on two rough inclined planes whose inclination on the horizontal is α, where, $w_1 \geq w_2$. These blocks are connected by a string that passes on a smooth roller at the highest level. If these blocks impending to move, prove that:

$$\tan\alpha = \mu(w_1 + w_2)/(w_1 - w_2),$$

where μ is the friction coefficient.

Solution:



From the equilibrium of the block w_1, we have:

$$R_1 = w_1 \cos\alpha$$

$$T = w_1 \sin\alpha - \mu R_1 = w_1 \sin\alpha - \mu w_1 \cos\alpha \qquad (1)$$

Similarly, from the equilibrium of the block w_2, we have:

$$R_2 = w_2 \cos\alpha$$

$$T = \mu R_2 + w_2 \sin\alpha = \mu w_2 \cos\alpha + w_2 \sin\alpha \qquad (2)$$

From eqs. (1) and (2) we note that:

$$w_1 \sin\alpha - \mu w_1 \cos\alpha = \mu w_2 \cos\alpha + w_2 \sin\alpha$$

$$(w_1 - w_2)\tan\alpha = \mu(w_1 + w_2)$$

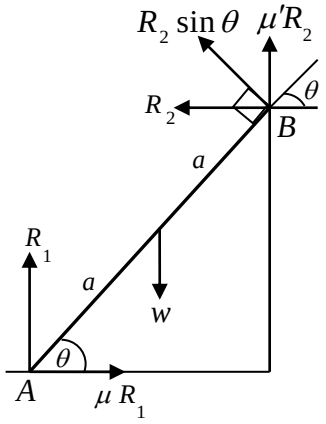
$$\therefore \tan\alpha = \mu(w_1 + w_2)/(w_1 - w_2)$$

Example3: A regular rod resting with its end A on a horizontal rough plane and with its end B on a vertical rough wall. If the rod is impending to slide, prove that:

$$\tan\theta = (1-\mu\mu')/2\mu$$

Where, θ is the angle between the rod and the horizontal plane, μ is the friction coefficient between the rod and the horizontal rough plane and μ' is the friction coefficient between the rod and the wall.

Solution:



From the equilibrium for the rod, we have:

$$R_1 = w - \mu' R_2 \qquad (1)$$

$$R_2 = \mu R_1 \qquad (2)$$

From eqs. (1) and (2) we note that:

$$R_1 = w - \mu\mu' R_1$$

$$\therefore R_1 = w/(1+\mu\mu') \qquad (3)$$

$$\therefore R_2 = \mu w/(1+\mu\mu') \qquad (4)$$

By taking the moment about the point A we have:

$$wa\cos\theta = 2aR_2\sin\theta + \mu' R_2 \cdot 2a\cos\theta$$

$$(w - 2\mu' R_2)\cos\theta = 2R_2\sin\theta$$

$$\therefore \tan\theta = \frac{w - 2\mu' R_2}{2R_2} = \frac{w - \dfrac{2\mu\mu' w}{(1+\mu\mu')}}{\dfrac{2\mu w}{(1+\mu\mu')}}$$

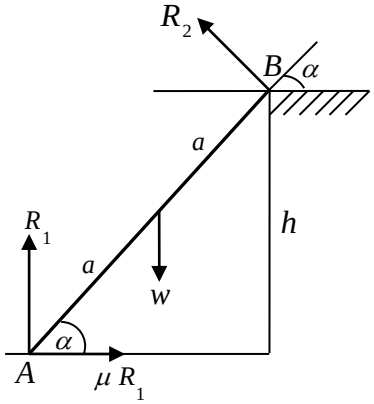
$$= \frac{1+\mu\mu' - 2\mu\mu'}{2\mu} = \frac{1-\mu\mu'}{2\mu}$$

Example4: A regular rod in an equilibrium state with a weight w resting with its bottom end on a rough horizontal plane. Also, it resting on a smooth horizontal barrier with a height h from the horizontal plane at another point of it, such that the vertical plane passes through the rod is perpendicular on the barrier. Prove that the friction coefficient between the rod and the plane is equal to:

$$2\sin^2\alpha\cos\alpha/(h - a\sin\alpha\cos^2\alpha)$$

Where α is the angle between the rod and the horizontal plane at a final equilibrium state.

Solution:



From this figure, we observe that:

$$AB = \frac{h}{\sin\alpha}$$

Also, from the equilibrium for the rod we have:

$$R_1 + R_2\cos\alpha = w \tag{1}$$

$$\mu R_1 = R_2\sin\alpha \tag{2}$$

$$\therefore R_1 = \frac{R_2}{\mu}\sin\alpha \tag{3}$$

Then

$$\frac{R_2}{\mu}\sin\alpha + R_2\cos\alpha = w$$

$$\therefore R_2 = \frac{\mu w}{\sin\alpha + \mu\cos\alpha} \tag{4}$$

By taking the moment about the point A we have:

$$wa\cos = \frac{R_2 h}{\sin\alpha} \tag{5}$$

Then

$$wa\cos\alpha = \frac{\mu w}{\sin\alpha + \mu\cos\alpha}\cdot\frac{h}{\sin\alpha}$$

$$\therefore a\cos\alpha\sin\alpha(\sin\alpha + \mu\cos\alpha) = \mu h$$

$$\therefore \mu(h - a\sin\alpha\cos^2\alpha) = a\cos\alpha\sin^2\alpha$$

$$\therefore \mu = \frac{a\cos\alpha\sin^2\alpha}{(h - a\sin\alpha\cos^2\alpha)}$$

Slipping and Tipping

Up to this point in our discussion of problems with friction, impending motion in a given system resulted from applied forces that create slipping at certain contacting surfaces. Recall that impending motion occurs at a surface where the friction force f and the normal force N are related by:

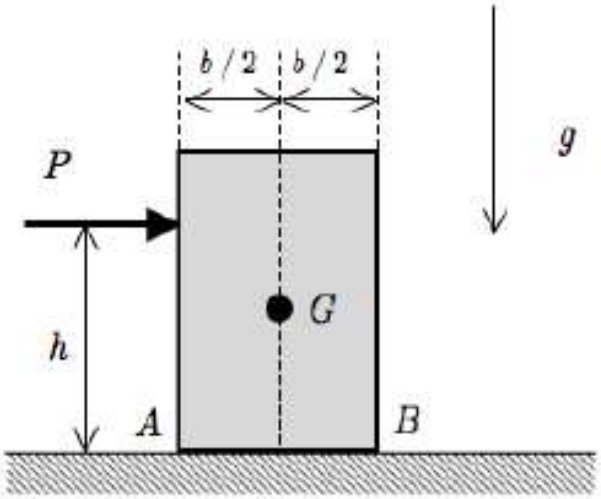
$$F_m = \mu_s N$$

In this section, we will study a different kind of problem where impending motion in frictional systems can result from either slipping (as described above) or from a pair of interfacing surfaces losing contact with each other. This second situation will be referred to as “tipping”.

As an introduction to this topic, consider the case shown below of a block of weight W and center of mass at G that is resting on a rough horizontal surface (with coefficient of static friction μ_s). A horizontal force P is applied to the block at vertical distance of h above the rough surface.

In the absence of the applied force P the normal reaction force on the block from the ground is immediately below the center of mass G, as shown below left. However, with the application of the force P, the point of application for the normal and friction forces has moved a distance of d to the right at point

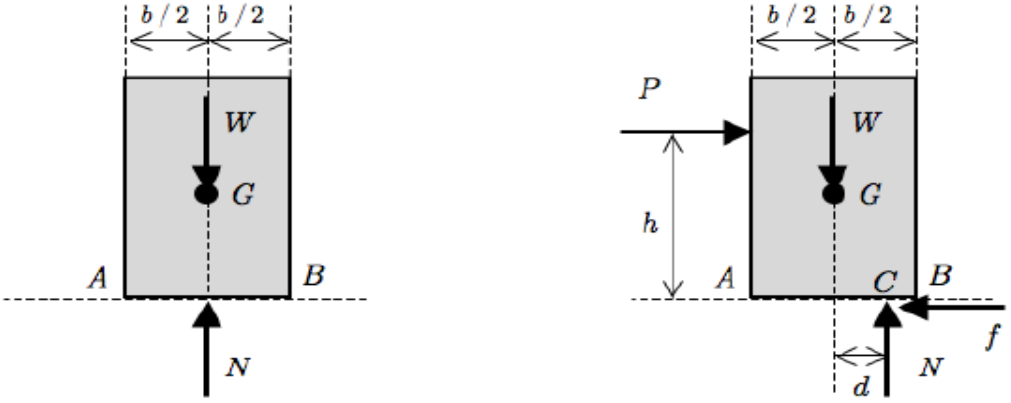
C, as shown in the following figure below right. How far to the right have these forces moved? That is, what is the value for d? To see this, we will write down a moment equilibrium equation about point C:



$$\sum M_C = Wd - Ph = 0 \qquad \Rightarrow \qquad d = \frac{P}{W}h$$

From this we see that these contact forces move to the right as either P or h is increased. When P = 0, d = 0 and the normal force N is directly below G, as described earlier.

What happens when this distance d falls outside of the contact region of the block with the ground (that is, when d>b/2)? The answer is that the block can no longer remain in equilibrium in rotation; it has "tipped" about corner B of the block.



Let us now look at the force equilibrium equations for the block:

$$\sum F_x = P - f = 0 \qquad \Rightarrow \qquad P = f$$

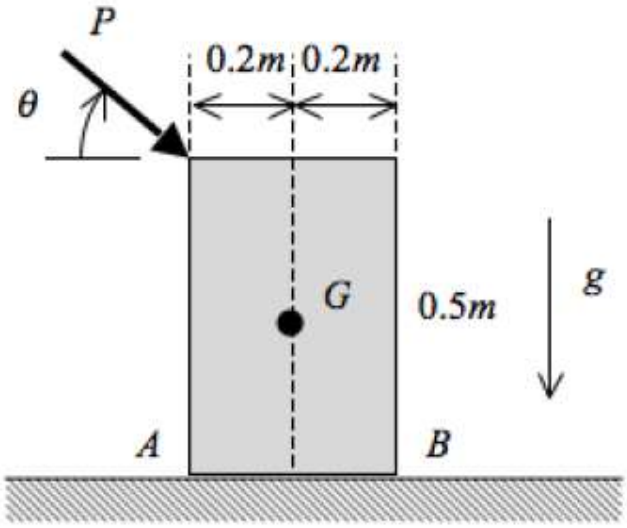
$$\sum F_y = N - W = 0 \qquad \Rightarrow \qquad N = W$$

From this we see that the block can no longer remain in equilibrium if P = f > $\mu_s (= \mu_s W)$ since, for this case, the block will slip on the horizontal surface.

We are now faced with the question as to whether the block will tend to tip or to slip first as the applied load P is increased. How do we answer that question?

There are many ways to deal with impending motion problems; those that require you to determine whether an object slips or tips. Here we will illustrate two of them.

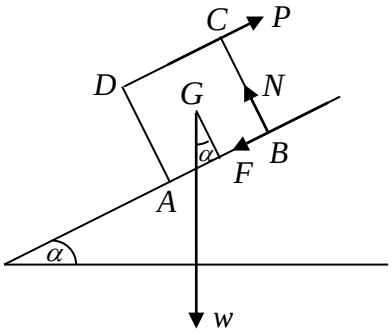
Example: The block has a weight of W=200N. The coefficient of static friction between the block and the rough horizontal surface is known to be $\mu_s = 0.5$.



1. If $\theta = 0$, what is the minimum force P required to initiate movement of the block? Will the resulting motion be tipping or slipping?
2. If $\theta = 40$, what is the minimum force P required to initiate movement of the block? Will the resulting motion be tipping or slipping?

Example: A square lamina *ABCD* is placed at a vertical level and it resting by its edge *AB* on a rough inclined plane that directed to the highest level. An increasing force affected the line *CD*. Prove that the lamina is impending to sliding or to tipping where the inclination angle of the plane on the horizontal is greater than or less than $\tan^{-1}(1-2\mu)$, where μ is the friction coefficient.

Solution:



Suppose that the lamina is impending to sliding when the inclination for the plane is α. The forces which affect the lamina are:

1- The lamina weight

2- The force P in the direction of DC

3- The reaction force N

4- The friction force F at B under the condition $F < \mu N$.

Then, from the equilibrium of the lamina we have:

$$P - F - w\sin\alpha = 0$$

$$N - w\cos\alpha = 0$$

Also, by taking the moment about the point B we have:

$$P \cdot 2a = w \cdot a\cos\alpha + w \cdot a\sin\alpha$$

Where $2a$ is the length of the edge of the lamina. So, from the previous equations we observe that:

$$F = \frac{1}{2} w(\cos\alpha - \sin\alpha)$$

$$N = w\cos\alpha$$

The sliding may be occurring if the condition $F < \mu N$ is satisfied. Then:

$$\frac{1}{2} w(\cos\alpha - \sin\alpha) < \mu w \cos\alpha$$

So,

$$1 - \tan\alpha < 2\mu \Rightarrow \therefore \tan\alpha > 1 - 2\mu$$

This is condition for the sliding process otherwise the tipping process may be occurred.

Problems

1- Pushing a block up a rough inclined plane. What force F is necessary to push a 5.00-kg block up a rough inclined plane at a constant velocity?

2- Using a horizontal force of 40 Newton, how heavy a box could you drag across a floor where the coefficient of friction is 0.40?

3- A uniform cylinder rests on a horizontal plane with its axis horizontal. A plank lies across the cylinder with one end resting on the plane. Show that if the plank is just about to slide on the cylinder the angle of friction between them is half the inclination of the plank to the horizontal.

4- A heavy cubical block of edge 2a is placed on a rough table with one face parallel to the edge of the table and at a distance a cot a from it; to the centre of this face a light smooth rod of length l is freely jointed, it passes over the smooth edge of the table and carries a weight W at its end. Show that as W is increased the equilibrium of the block is broken by its tilting about an edge if

$$\mu > \frac{l\cos\alpha\sin^2\alpha}{a + l\sin\alpha\cos\alpha(\sin\alpha - \cos\alpha)}$$

5- A uniform circular cylindrical log of radius a and weight W lies with its axis horizontal between two rough parallel horizontal rails at the same level and at a distance $2a\sin\alpha$ apart; show that, if a gradually increasing couple be applied to the log in a plane perpendicular to the rails and axis, the log will turn over one of the rails when the couple is of magnitude $Wa\sin\alpha$, provided the angle of friction λ is greater than α, but otherwise the log will turn about the axis when the couple is

$$Wa\sin\lambda\cos\lambda\sec\alpha$$

6- A particle rests on a rough horizontal plane under the action of a horizontal force P. A gradually increasing horizontal force is applied to it perpendicular to P. Determine the direction in which the particle will begin to move.

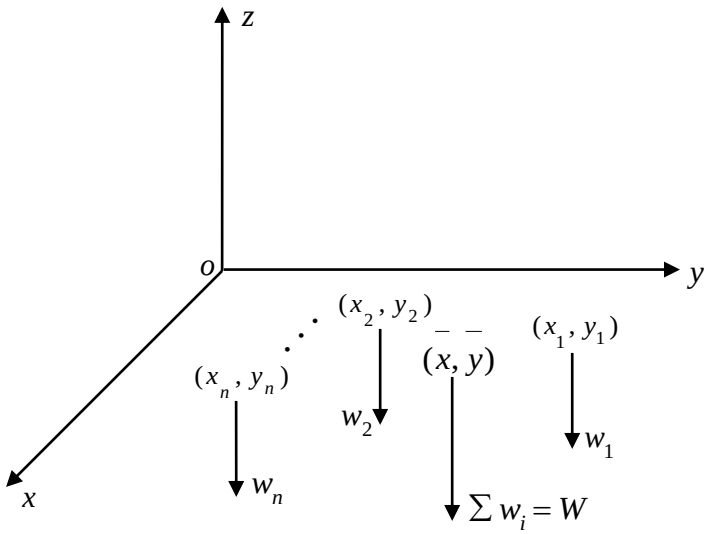
7- Find the magnitude and direction of application of the least force necessary to drag a particle up a rough inclined plane.

8- A circular hoop hangs over a horizontal peg and a weight 1.5 times that of the hoop hangs tangentially from it. Show that if the hoop is about to slip on the peg the coefficient of friction is 0.75.

Chapter 5
Center of Mass

Center of Mass

The center of mass (C.M.) is the point where the total mass of the object is visually concentrated. It is again determined from the principle of moment. C.M. is independent of the gravitational effect. It is a unique property of the object, depending on its shape and material. For simplification in the derivation of the C.M., apply the same assumptions as above; uniform intensity of the earth's force field over the body and parallel field of force due to the gravitational attraction.



To deal with the situation of n masses, $w_1, w_2, \ldots, w_n$ located at coordinate positions $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, where

$$\sum_{i=1}^{n} w_i = w_1 + w_2 + \cdots + w_n = W$$

And

$$\therefore w_1 x_1 + w_2 x_2 + \cdots + w_n x_n = W\bar{x}$$

Then

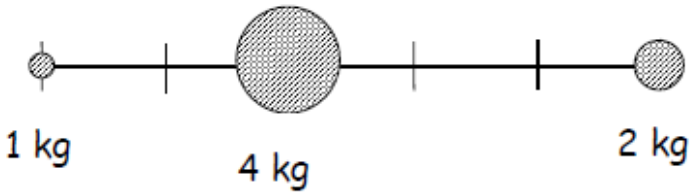
$$\therefore \bar{x} = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Similarly

$$\bar{y} = \frac{\sum_i w_i y_i}{\sum_i w_i}$$

We must observe that these previous laws are established for separate points or simple geometric shapes of constant density.

Example1: Find the center of mass of three particles:



Solution:

$$x_{cm} = \frac{(1kg)(0m) + (4kg)(2m) + (2kg)(5m)}{1kg + 4kg + 2kg} = \frac{18}{7} m = 2.57m$$

Example2: Masses of 5kg, 3kg and 9kg are located at the points with coordinates (−1, 1), (4, 3), and (8, 7) respectively. Find the coordinates of their centre of mass.

Solution: Here n = 3.

$$\bar{x} = \frac{\sum_{i=1}^{3} m_i x_i}{\sum_{i=1}^{3} m_i} = \frac{5(-1) + 3(4) + 9(8)}{5 + 3 + 9} = \frac{79}{17} = 4.65$$

Hence the centre of mass is located at the point (4.65, 4.53).

Center of Mass for some Continuous Particles

In center of mass calculations, (something) is position, but we will evaluate similar integrals where (something) is some other scalar or vector function of position. Most often we label the material by its spatial position, and evaluate dm in terms of increments of position.

Using the integration to get the center of mass

Similarly for the previous section, the coordinate positions for the central of mass take the form:

$$\overline{x_1} = \frac{\sum \rho x \Delta s}{\sum \rho \Delta s} \quad , \quad \overline{y_1} = \frac{\sum \rho y \Delta s}{\sum \rho \Delta s}$$

Where, Δs is the particle mass or particle volume, ρ is the density of the particle and $(\overline{x_1}, \overline{y_1})$ is the centre of mass for the particle. Also, the coordinate positions for the total central of mass is $(\overline{x}, \overline{y})$, where

$$\overline{x} = \lim_{\Delta s \to 0} \overline{x_1} \quad , \quad \overline{y} = \lim_{\Delta s \to 0} \overline{y_1}$$

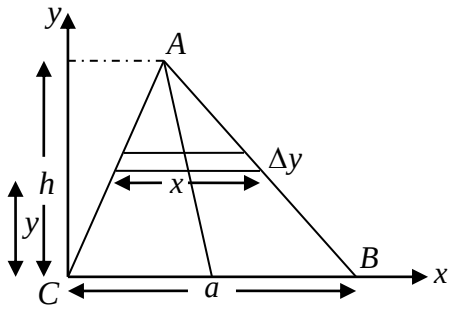
$$\bar{x} = \frac{\lim_{\Delta s \to 0} \sum \rho\, x \Delta s}{\lim_{\Delta s \to 0} \sum \rho\, \Delta s} = \frac{\int \rho\, x\, ds}{\int \rho\, ds} = \frac{\int x\, dw}{\int dw}$$

$$\bar{y} = \frac{\lim_{\Delta s \to 0} \sum \rho\, y \Delta s}{\lim_{\Delta s \to 0} \sum \rho\, \Delta s} = \frac{\int \rho\, y\, ds}{\int \rho\, ds} = \frac{\int y\, dw}{\int dw}$$

Center of mass for some special cases

1- Center of mass for a lamina in the form of triangle

Suppose we take a slice from a triangle in the form of rectangle with length x and width Δy as illustrated from the following figure. Then, this slice has a weight $\rho x \Delta y$, where ρ is a density of the triangle. Let $(\bar{x}, \bar{y})$ is the coordinates for the central of mass for the triangle.



$$\because \bar{y} = \frac{\int y \rho\, x\, dy}{\int \rho\, x\, dy}$$

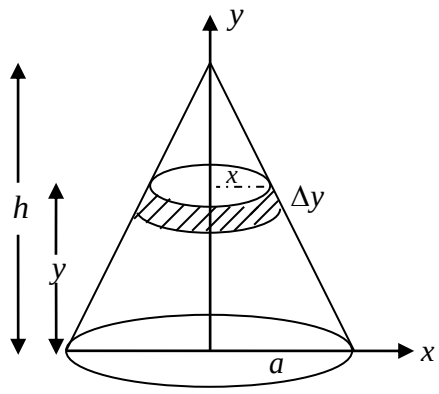
But from the figure we have:

$$\frac{x}{a} = \frac{h-y}{h} \Rightarrow \therefore x = \frac{a(h-y)}{h}$$

$$\therefore \bar{y} = \frac{\int_0^h y(h-y)dy}{\int_0^h (h-y)dy} = \frac{\left|\frac{hy^2}{2} - \frac{y^3}{3}\right|_0^h}{\left|hy - \frac{y^2}{2}\right|_0^h} = \frac{h^3\left(\frac{1}{2} - \frac{1}{3}\right)}{h^2\left(1 - \frac{1}{2}\right)} = \frac{h}{3}$$

2- **Center of mass for solid circular cone**

Let we take a slice from a cone in the form of cylinder with radius x and height Δy as illustrated from the following figure. Then, this slice has a weight $\rho\pi x^2 \Delta y$, where ρ is a density of the cone. Also, from the figure we have:



$$\frac{x}{a} = \frac{h-y}{h} \Rightarrow \therefore x = \frac{a(h-y)}{h}$$

Then the slice weight becomes:

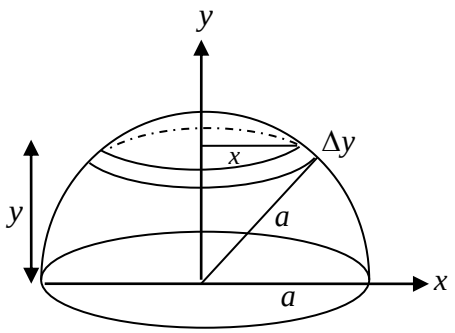
$$\rho\pi \frac{a^2}{h^2}(h-y)^2 \Delta y$$

Also, the central of mass for the slice is $(0, y)$ as illustrated from the figure. So,

$$\bar{x}=0 \quad , \quad \bar{y}=\frac{\int_0^h \rho\pi \frac{a^2}{h^2}(h-y)^2 y dy}{\int_0^h \rho\pi \frac{a^2}{h^2}(h-y)^2 dy}$$

$$\therefore \bar{y}=\frac{\int_0^h (h^2y-2hy^2+y^3)ydy}{\int_0^h (h^2-2hy+y^2)dy}=\frac{h^4\left(\frac{1}{2}-\frac{2}{3}+\frac{1}{4}\right)}{h^3\left(1-1+\frac{1}{3}\right)}=\frac{h}{4}$$

3- **Center of mass for solid semicircular lamina of radius a**



Let we take a slice from a semicircular lamina in the form of cylinder with radius x and height Δy as illustrated from the following figure. Then, this slice has a weight $\rho\pi x^2\Delta y$, where ρ is a density of the semicircular lamina. Also, from the figure we have:

$$x^2+y^2=a^2$$

And

$$\delta w=\rho\pi x^2\Delta y=\rho\pi(a^2-y^2)\Delta y$$

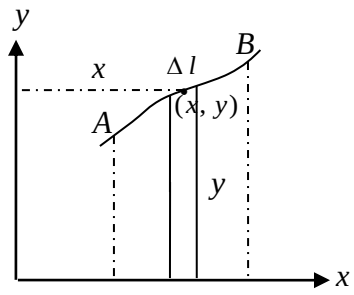
Also, the central of mass for the slice is $(0,y)$ as illustrated from the figure. So,

$$(\bar{x},\bar{y})=(0,\bar{y})$$

$$\therefore \bar{y} = \frac{\int_0^a \rho\pi(a^2 - y^2)ydy}{\int_0^a \rho\pi(a^2 - y^2)dy} = \frac{3}{8}a$$

Center of mass for any curve

To determine a center of mass for any curve $y = f(x)$, we firstly assume that any element from the curve with a length dl. So, this element has a weight $dw = \rho\, dl$ and it's central of mass is (x, y).



Then

$$\therefore \bar{x} = \frac{\lim_{\Delta l \to 0} \sum \rho\, x \Delta l}{\lim_{\Delta l \to 0} \sum \rho\, \Delta l} = \frac{\int \rho\, x\, dl}{\int \rho\, dl}$$

Similarly,

$$\bar{y} = \frac{\int \rho\, y\, dl}{\int \rho\, dl}$$

Also, we know that:

$$(dl)^2 = (dx)^2 + (dy)^2$$

$$\Rightarrow \frac{dl}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \quad , \quad \frac{dl}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

In the same context, if the equation of the curve is in the polar form (r,θ), then the length elements are $(dr, rd\theta)$ and the central of mass for the element is $(r\cos\theta, r\sin\theta)$. So,

$$dl = \sqrt{1+r^2\left(\frac{d\theta}{dr}\right)^2}\, dr$$

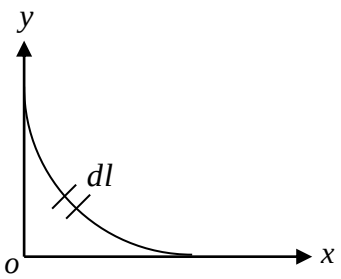
Where,

$$(dl)^2 = (dr)^2 + (rd\theta)^2$$

Example3: Find the center of mass for the curve $x^{2/3} + y^{2/3} = a^{2/3}$ which appears in the first quarter.

Solution:

Let the element from this curve has a length dl and it's central of mass is (x, y)



Where,

$$(dl)^2 = (dx)^2 + (dy)^2$$

$$\therefore dl = \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx = \left(\frac{a}{x}\right)^{1/3} dx$$

and

$$dw = \rho\, dl = \rho\left(\frac{a}{x}\right)^{1/3} dx$$

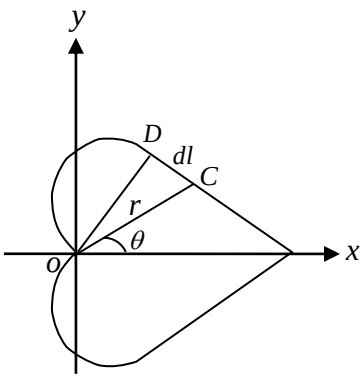
$$\therefore \bar{x} = \frac{\int x dw}{\int dw} = \frac{\int_0^a \rho\, x \cdot \left(\frac{a}{x}\right)^{1/3} dx}{\int_0^a \rho \left(\frac{a}{x}\right)^{1/3} dx} = \frac{2}{5}a$$

$$\therefore \bar{x} = \bar{y} = \frac{2}{5}a$$

Example4: Find the center of mass for the curve $r = a(1+\cos\theta)$, $-\pi \le \theta \le \pi$.

Solution:

Let DC has an element length dl as illustrated in the following figure. Also the polar coordinates (r,θ) for the point C are $x = r\cos\theta$, $y = r\sin\theta$.



Then from the equation of the curve we note that $\frac{dr}{d\theta} = -a\sin\theta$

So, we observe that

$$\therefore dl = \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2}\, d\theta = \sqrt{a^2 \sin^2\theta + r^2}\, d\theta$$

$$= \sqrt{a^2 \sin^2\theta + a^2 + a^2 \cos^2\theta + 2a^2 \cos\theta}\, d\theta$$

$$= \sqrt{2a^2 + 2a^2 \cos\theta}\, d\theta$$

$$= \sqrt{2} \cdot a\sqrt{1+\cos\theta}\, d\theta = a\sqrt{2} \cdot \sqrt{2} \cos\frac{\theta}{2}\, d\theta$$

$$= 2a \cos\frac{\theta}{2}\, d\theta$$

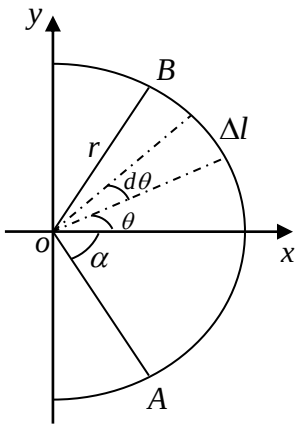
From the similarity for the curve about the x-axis, then we must note that $\bar{y} = 0$. Likewise, suppose that $(\bar{x},0)$ is the coordinate of the center of mass. Then,

$$\therefore \bar{x} = \frac{\int \rho x\, dl}{\int \rho\, dl} = \frac{\int x\, dl}{\int dl} = \frac{\int_{-\pi}^{\pi} r\cos\theta \cdot 2a\cos\frac{\theta}{2}\, d\theta}{\int_{-\pi}^{\pi} 2a\cos\frac{\theta}{2}\, d\theta} = \frac{4a}{5}$$

Example5: Find the center of mass for a circular arc which has a central angle 2α.

Solution:

Suppose that the circular arc is similar about the x-axis as illustrated from the following figure. Also, let the element from this curve has a length dl, a weight $\rho\, dl$ and it's central of mass is (x, y).



So,

$$\bar{x} = \frac{\int \rho x \, dl}{\int \rho \, dl} \quad , \quad \bar{y} = 0$$

After using the polar coordinates, then we have:

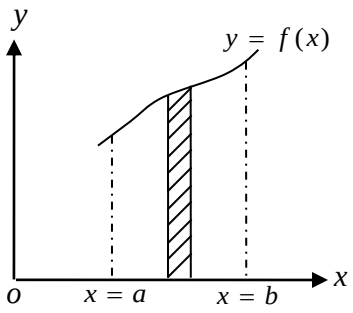
$$\therefore \bar{x} = \frac{\int_{-\alpha}^{\alpha} r^2 \cos\theta \, d\theta}{\int_{-\alpha}^{\alpha} r \, d\theta} = \frac{r \int_{-\alpha}^{\alpha} \cos\theta \, d\theta}{\int_{-\alpha}^{\alpha} d\theta} = \frac{r \cdot \sin\theta \big|_{-\alpha}^{\alpha}}{\theta \big|_{-\alpha}^{\alpha}}$$

$$= \frac{r(\sin\alpha - \sin(-\alpha))}{\alpha - (-)} \alpha = \frac{r \sin\alpha}{\alpha}$$

Center of mass for any area

Firstly, we must mention here that the appropriate coordinates for calculating the center of mass for any area must be chosen. So, it is advised to choose the coordinate system describing the area as appropriate for the particular shape of the object. If the body is of rectangular shape, we should use rectangular coordinates. However, if its shape is circular, we should employ polar coordinates instead. Then,

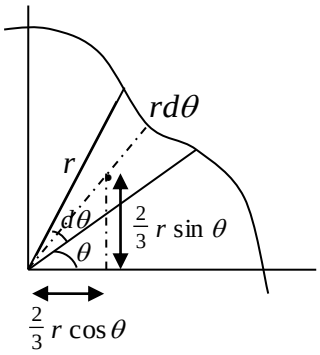
we assume the thickness is constant over the entire area. To calculate the centre of the area which bounded by the curve $y = f(x)$ and the lines $x = a$,as illustrated in the $x = b$ following diagram.



Let the slice which parallel to the y-axes has a weight $dw = \rho\, y dx$ and a center of mass $(x, y/2)$. So, we note that

$$\therefore \bar{x} = \frac{\int_a^b \rho\, yx dx}{\int_a^b \rho\, y dx} \quad , \quad \bar{y} = \frac{\int_a^b \rho\, y \cdot (y/2) dx}{\int_a^b \rho\, y dx}$$

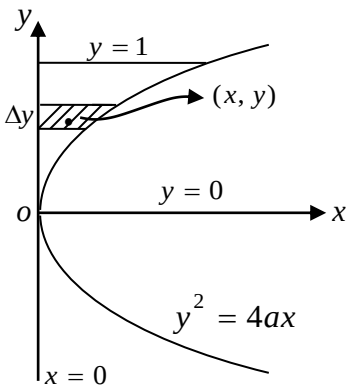
Here, we must refer that if the area is located by a polar curve in the form $r = f(\theta)$, then we take a slice with area equal to the area of the triangle in the following figure which is clearly equal to $d\sigma = \frac{1}{2} r d\theta \cdot r$. Also, this triangle slice has a weight $dw = \frac{\rho}{2} r^2 d\theta$ and a center of mass $\left(\frac{2}{3} r\cos\theta, \frac{2}{3} r\sin\theta\right)$ as illustrated in the following figure.



Example6: Find the center of mass for the area which is bounded by the curve $y^2 = 4ax$, the y-axes and the two lines $y = 0, y = 1$.

Solution:

Let we take the slice from the area which parallel to the x-axes. As we know this slice with width Δy and a length x has a weight $dw = \rho x dy$ and a center of mass $\left(\frac{x}{2}, y\right)$. So, we note that:



$$\bar{x} = \frac{\int_0^1 \rho x \cdot \frac{x}{2} dy}{\int_0^1 \rho x dy} = \frac{1}{2} \cdot \frac{\int_0^1 x^2 dy}{\int_0^1 x dy}$$

Also, from the equation of the curve, we note that $x = y^2 / 4a$, then:

$$\bar{x} = \frac{1}{2} \cdot \frac{\int_0^1 \frac{y^4}{16a^2} dy}{\int_0^1 \frac{y^2}{4a} dy} = \frac{1}{8a} \frac{\left.\frac{y^5}{5}\right|_0^1}{\left.\frac{y^3}{3}\right|_0^1} = \frac{3a}{40}$$

Similarly,

$$\bar{y} = \frac{\int_0^1 \rho x \cdot y dy}{\int_0^1 \rho x dy} = \frac{\int_0^1 \frac{y^3}{4a} dy}{\int_0^1 \frac{y^2}{4a} dy} = \frac{3a}{4}$$

Then, the center of mass for this area is $\left(\frac{3a}{40}, \frac{3a}{4}\right)$.

Example7: Find the center of mass for the area which is bounded by the two curves $y^2 = 4ax$, $x^2 = 4ay$.

Solution:

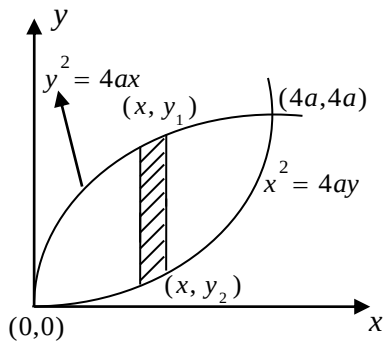
First we find the intersection points of the two curves $y^2 = 4ax$, $x^2 = 4ay$ as follows:

$$\therefore y^2 = 4a \cdot 2a^{1/2} y^{1/2} = 8a^{3/2} y^{1/2}$$
$$\therefore y^2 - 8a^{3/2} y^{1/2} = 0 \Rightarrow y^{1/2}(y^{3/2} - 8a^{3/2}) = 0$$
$$\therefore y = 0 \quad or \quad y^{3/2} = 8a^{3/2}$$

Then

$$y^3 = 64a^3 \Rightarrow \therefore y = 4a \Rightarrow \therefore x = 4a$$

Then, the intersection points are $(0,0),(4a,4a)$. Here, let we take a slice which parallel to the y-axes with width Δx and a measured length between the two points $(x,y_1),(x,y_2)$. So, the area of this slice is $(y_1-y_2)\Delta x$, its weight $\rho(y_1-y_2)\Delta x$ and a center of mass $\left(x,\frac{y_1+y_2}{2}\right)$. So, we note from the following figure that the point (x,y_1) satisfy the curve $y^2=4ax$ and the point (x,y_2) satisfy the curve $x^2=4ay$. Then,



The weight of the slice becomes

$$dw=\rho(2\sqrt{ax}-\frac{x^2}{4a})dx$$

And its centre of mass is the point

$$\left(x,\frac{1}{2}\left(2\sqrt{ax}+\frac{x^2}{4a}\right)\right)$$

$$\therefore \bar{x}=\frac{\int_0^{4a}\rho\left(2\sqrt{ax}-\frac{x^2}{4a}\right)xdx}{\int_0^{4a}\rho\left(2\sqrt{ax}-\frac{x^2}{4a}\right)dx}=\frac{9a}{5}$$

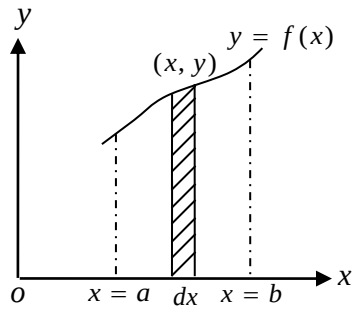
Similarly,

$$\bar{y} = \frac{1}{2} \cdot \frac{\int_0^{4a} \rho\left(2\sqrt{ax} - \frac{x^2}{4a}\right)\left(2\sqrt{ax} + \frac{x^2}{4a}\right)dx}{\int_0^{4a} \rho\left(2\sqrt{ax} - \frac{x^2}{4a}\right)dx} = \frac{9a}{5}$$

Then, the center of mass for this area is $\left(\frac{9a}{5}, \frac{9a}{5}\right)$.

Center of mass of rotational volumes

To find the centre of mass of a volume of revolution, i.e. a solid whose boundary is obtained by revolving a given curve $y = f(x)$ round the axis of x, cut off between plane ends x=a, x=b. Since sections at right angles to the axis of x are circles, the solid may be regarded as built up of thin discs of area πy^2 and thickness dx, and the x-coordinate of the centre of mass is given by

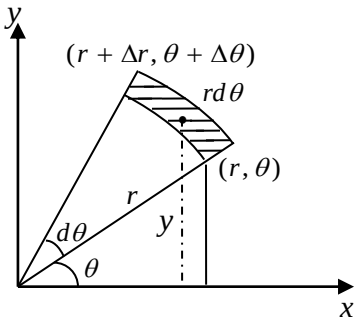


So,

$$\bar{x} = \frac{\int \rho\pi\, y^2 x dx}{\int \rho\pi\, y^2 dx}$$

Center of mass of rotational volumes via polar coordinates

Suppose that an element in a form of rectangle with area $rdrd\theta$ as illustrated in the following figure.



If this element rotated about the x-axis. So, the resultant volume becomes $dv = 2\pi r \sin\theta \cdot rdrd\theta$ and it weighted $dw = 2\pi \rho r^2 \sin\theta \, drd\theta$. So,

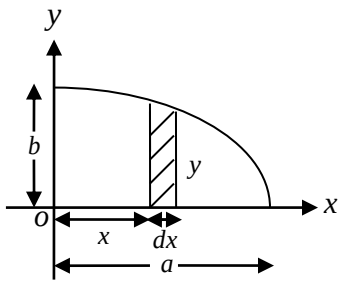
$$\bar{x} = \frac{\int x\, dw}{\int dw}$$

Example8: Find the center of mass for the volume which results from the rotation of the area bounded by the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the y-axis about the x-axis.

Solution:

From the following diagram we note that the rotating disk about ox axis with a length y and a width dx has an area equal to πy^2 and a volume $dv = \pi y^2 dx$, then the weight for the disk

is $dw = \rho\pi\, y^2 dx$. Also, the center of mass for this volume is $(\bar{x},0)$, where



$$\bar{x} = \frac{\int_0^a \rho\pi\, y^2 x dx}{\int_0^a \rho\pi\, y^2 dx}$$

But

$$y^2 = b^2\left(1-\frac{x^2}{a^2}\right)$$

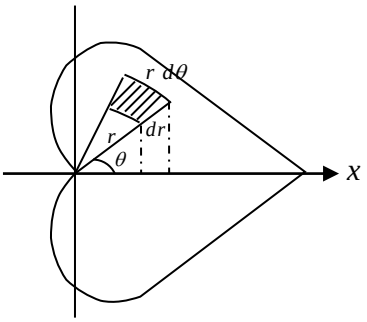
$$\therefore \bar{x} = \frac{\int_0^a x\left(1-\frac{x^2}{a^2}\right)dx}{\int_0^a \left(1-\frac{x^2}{a^2}\right)dx} = \frac{\left(\frac{x^2}{2}-\frac{x^4}{4a^2}\right)_0^a}{\left(x-\frac{x^3}{3a^2}\right)_0^a} = \frac{a^2/4}{2a/3} = \frac{3a}{4}$$

Example9: Find the center of mass for the volume which results from the rotation of the area bounded by the curve $r = a(1+\cos\theta)$ a full cycle about its axis.

Solution:

From the following diagram we note that the element with a length $rd\theta$ and a width dr has an area equal to $rd\theta\, dr$ and a volume $dv = 2\pi r\sin\theta \cdot rdrd\theta$, then the weight for this

element $dw = 2\pi\rho r^2 \sin\theta\, dr d\theta$. Also, the center of mass for this volume is $(\bar{x},0)$, where



$$\bar{x} = \frac{\int_0^{\pi}\int_0^{a(1+\cos\theta)} 2\pi\rho r^2 \sin\theta \cdot r\cos\theta\, dr d\theta}{\int_0^{\pi}\int_0^{a(1+\cos\theta)} 2\pi\rho r^2 \sin\theta\, dr d\theta}$$

$$= \frac{\int_0^{\pi}\left(\frac{r^4}{4}\right)_0^{a(1+\cos\theta)} \sin\theta\cos\theta\, d\theta}{\int_0^{\pi}\left(\frac{r^3}{3}\right)_0^{a(1+\cos\theta)} \sin\theta\, d\theta}$$

$$= \frac{3}{4}a \cdot \frac{\int_0^{\pi}(1+\cos\theta)^4 \sin\theta\cos\theta\, d\theta}{\int_0^{\pi}(1+\cos\theta)^3 \sin\theta\, d\theta}$$

To calculate this integration, let

$$\cos\theta = y - 1 \quad \Rightarrow \quad \sin\theta\, d\theta = -dy$$

$$at\ \ \theta = 0 \Rightarrow \therefore\ y = 2$$

$$at\ \ \theta = \pi \Rightarrow \therefore\ y = 0$$

$$\Rightarrow \int_0^{\pi}(1+\cos\theta)^4 \sin\theta\cos\theta\, d\theta$$

$$= \int_2^0 y^4(1-y)dy = \frac{64}{15}$$

And

$$\int_0^{\pi}(1+\cos\theta)^3\sin\theta\,d\theta$$

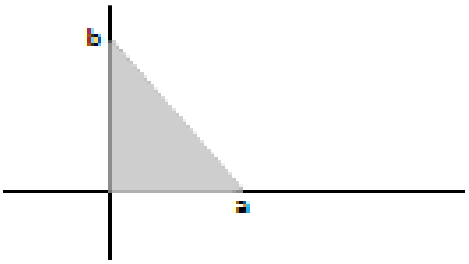
$$=-\frac{1}{4}\int_0^{\pi}(1+\cos\theta)^3\,d(1+\cos\theta)$$

$$=(1+\cos\theta)^3\Big|_0^{\pi}=4$$

$$\Rightarrow \bar{x}=\frac{3}{4}a\cdot\frac{64}{15\times 4}=\frac{4a}{5}$$

Exercise

1. Find the center of mass of the triangular region



2. Find the center of mass for each of the following

(i) The Parabola $y^2 = 4ax$, the x-axes and the straight line $x = b$.

(ii) The Parabola $\sqrt{\frac{x}{a}} + \sqrt{\frac{b}{y}} = 1$, the axial coordinates and the two straight lines $x = a, x = b$.

(iii) The Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, which is bounded by the positive the axial coordinates.

3- Find the center of mass of the top half of the unit disc.

4- Find the center of mass of the top half of an ellipse.

5. Find the center of mass for the area which bounded between:

(i) The curve $y = \sin\theta$ and the lines $x = 0, x = \pi/2$.

(ii) The curve $r = a(1 - \cos\theta)$.

(iii) The Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, which is bounded by the positive the axial coordinates.

6. Find the center of mass for the volume which is results from the rotation the area which bounded by the curve $r = a(1+\cos\theta)$ about its axis.

7. Find the center of mass for the curve $y = a(1-\cos\theta)$, $x = a(\theta+\sin\theta)$ which is bounded by the first positive quarter.

8. Find the center of the volume which results from the rotation for the curve $x = a(\theta+\sin\theta)$, $y = a(1-\cos\theta)$ about the oy axis.

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